

A Project Report on

IoT-Based Smart Air Quality Monitoring and Purification System Using Plants

submitted by

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DECLARATION

We hereby proclaim that the work **IoT-Based Smart Air Quality Monitoring and Purification System Using Plants** has been presented in this report. Information derived from the other sources to complete this work has been accredited in the text and a list of references has been given.

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VERIFICATION

This is to verify that the project entitled **IoT-Based Smart Air Quality Monitoring and Purification System Using Plants** was submitted to the department of Electronics and communication engineering department of Bundelkhand Institute of Engineering and Technology, Jhansi.

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DEDICATION

To my beloved Parents and almighty

PROJECT OUTCOMES(Ps)

S. N.	Project outcomes After completing this project students will be able to	Bloom's knowledge level
P1	Apply theoretical knowledge to develop a practical solution for autonomous air quality monitoring system.	KL3
P2	Design and integrate multiple components, sensors, and actuators to create an efficient monitoring of indoor air and irrigation system for plant.	KL6
P3	Analyse the project requirements and constraints to effectively manage and execute the IoT-based air quality monitoring and purification system.	KL4

KL: Bloom's knowledge level, KL1: remember, KL2: Understand, KL3: Apply, KL4: Analyse, KL5: Evaluate, KL6: Create/Design

Mapping of project outcomes with Program Outcomes (POs)

S. N.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
P1	2	2	2	3	2	3	2	2	-	2	2	2	3	2
P2	2	3	3	3	2	3	3	3	-	3	2	3	3	3
P3	2	3	2	3	2	3	3	2	-	3	2	2	2	3

Mapping rules (Rubrics)

1: poor 2: medium 3: best

Mapping of Program Outcomes(POs) with Project

S. N.	Program outcomes After completing this project students will be able to	Map
PO1	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.	2
PO2	Problem Analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.	3
PO3	Design/development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.	3
PO4	Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.	2
PO5	Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern Engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations	2
PO6	The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.	2

PO7	Environment and Sustainability: Understand the impact of the professional Engineering solutions in societal and Environmental contexts, and demonstrate the knowledge of, and need for sustainable development.	3
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice.	3
PO9	Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams.	3
PO10	Communication: Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations. instructions.	2
PO11	Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	-
PO12	Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change	2
PSO1	Apply the fundamental concepts of Electronics and Communication Engineering to design a variety of components and systems for applications like Signal processing, Image Processing, Communication, Networking, Microcontrollers, Embedded systems, Control Systems.	2
PSO2	Apply the concepts learned in Electronics, Signal Processing, Image Processing, Communication, Networking, Microcontrollers, Embedded Systems, Control System to arrive at solutions to real-world problems.	2

Mapping rules (Rubrics)

1: poor 2: medium 3: best

ABSTRACT

Air pollution, especially in indoor environments, has become a major concern due to its direct impact on human health and comfort. Traditional air purifiers are often expensive and lack real-time monitoring capabilities. To address this problem, this project titled “**IoT-Based Smart Air Quality Monitoring and Purification System Using Plants**” presents a cost-effective and eco-friendly solution that combines modern IoT technology with the natural purifying properties of plants.

The system uses a set of air quality sensors—**MQ135** for detecting harmful gases like CO₂, NH₃, and benzene, **MQ9** for monitoring carbon monoxide and flammable gases, and **PM2.5 GP2Y1010AU0F** for measuring fine dust particles in the air. Alongside, a **DHT11** sensor is used to check temperature and humidity. All data is collected and processed using a **ESP32** microcontroller, which connects wirelessly to the **Blynk IoT app**, allowing users to monitor the air quality in real-time from their smartphones.

To improve air quality naturally, a **snake plant** is integrated into the system for its proven ability to absorb pollutants and produce oxygen. To ensure the plant remains healthy without manual effort, an **automatic irrigation system** is implemented using a **soil moisture sensor**, **relay module**, and a **mini water pump**. The system automatically waters the plant when the soil becomes dry, ensuring consistent care.

This project promotes the use of **renewable and natural solutions** alongside **smart electronics** to build a reliable and automated air monitoring and purification system. It is ideal for homes, offices, classrooms, or any small indoor environment where clean air is essential. By combining technology and nature, the system supports healthier living in a sustainable and user-friendly way.

KEYWORDS: Air Quality Monitoring, IoT, ESP32, MQ135, PM2.5 Sensor, Smart Irrigation, Snake Plant, Blynk App, Indoor Pollution, Air Purification System.

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CONTRIBUTIONS

The main objective of this project is to monitor air quality and soil moisture in real-time and improve air purification using plant-based and IoT-assisted techniques. The main contributions of the work are:

1. IoT Technology has been used for real-time air quality and soil moisture monitoring using ESP32 microcontroller and multiple sensors.
2. Prediction and automation are done using Blynk IoT platform with datastreams and mobile app-based monitoring and control.
3. Air quality assessment has been done by integrating multiple sensors to measure gases, particulate matter, temperature, and humidity.
4. Hybrid air purification has been implemented by combining plant-based purification with sensor-controlled ventilation systems to enhance air cleaning efficiency.
5. ESP32 and Blynk cloud have been integrated to transmit sensor data in real-time for user monitoring and automatic control of water pump and fans through mobile application.

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ABBREVIATIONS AND SYMBOLS

ABBREVIATIONS

IoT	Internet of Things
AQI	Air Quality Index
PM	Particulate Matter
VOC	Volatile Organic Compounds
ESP32	Espressif Systems Microcontroller (32-bit)
Wi-Fi	Wireless Fidelity
LCD	Liquid Crystal Display
OLED	Organic Light Emitting Diode
PCB	Printed Circuit Board
MQ	Metal Oxide Semiconductor (Gas Sensor Series)
DHT	Digital Humidity and Temperature Sensor
CO	Carbon Monoxide
CO ₂	Carbon Dioxide
NO ₂	Nitrogen Dioxide
BLE	Bluetooth Low Energy
Blynk	IoT Mobile Application Platform
DC	Direct Current
AC	Alternating Current
USB	Universal Serial Bus
LED	Light Emitting Diode
API	Application Programming Interface
ML	Machine Learning
RH	Relative Humidity

SYMBOLS

$\mu\text{g}/\text{m}^3$	Micrograms per cubic meter (used for PM levels)
$^{\circ}\text{C}$	Degree Celsius (temperature)
%	Percent (humidity, soil moisture)
V	Voltage
A	Ampere (current)
Ω	Ohm (resistance)
mA	Milliampere
$\text{k}\Omega$	Kilo Ohm
ppm	Parts per million (gas concentration)
ug/m^3	Microgram per cubic meter (particulate concentration)

CHAPTER 1

INTRODUCTION

1.1 Background of Air Pollution

Air pollution has emerged as one of the most pressing environmental challenges in the modern world. Rapid industrialization, urbanization, vehicular emissions, construction activities, and deforestation have led to an alarming rise in the concentration of pollutants in the atmosphere. Major air pollutants include particulate matter (PM_{2.5} and PM₁₀), nitrogen oxides (NO_x), sulfur dioxide (SO₂), carbon monoxide (CO), and volatile organic compounds (VOCs). Among these, fine particulate matter such as PM_{2.5} is particularly dangerous as it penetrates deep into the lungs and even enters the bloodstream, causing serious health issues [3].

According to the World Health Organization (WHO), air pollution is responsible for approximately 7 million premature deaths each year globally [3]. Countries like India, China, and parts of Southeast Asia consistently record air quality indices (AQI) in the 'unhealthy' or 'hazardous' range, especially during winter months due to seasonal burning of crops and temperature inversions [5]. Indoor air pollution is another hidden threat caused by cooking, smoking, and the use of chemical products inside closed spaces.

Traditional methods to combat air pollution include emission control regulations and large-scale green cover initiatives. However, with technological advancement, newer solutions such as air purifiers, real-time air quality monitoring, and IoT-based systems are gaining attention. There is an urgent need to develop sustainable and intelligent solutions that not only monitor but also actively purify the air we breathe [8].

1.2 Need for Air Quality Monitoring

Accurate and real-time air quality monitoring is the first step toward addressing the problem of air pollution. Most cities rely on fixed monitoring stations operated by government agencies; however, these are limited in number and often fail to represent localized pollution levels experienced in specific areas. For instance, pollution levels inside a school located near a busy road could be significantly higher than what is reported for the city as a whole [35].

Personal and localized monitoring systems enable individuals and communities to become aware of the air quality around them. This awareness is critical in taking preventive measures such as wearing masks, using air purifiers, and avoiding outdoor activities during high pollution episodes.

Moreover, industries and large commercial spaces are now required to monitor their emissions and maintain indoor air quality to comply with environmental regulations. Real-time data also helps policymakers and researchers identify pollution hotspots and develop effective mitigation strategies [28].

With the advent of low-cost sensors and the Internet of Things (IoT), it is now feasible to build scalable air quality monitoring networks that provide granular, real-time data. Such systems empower not just governments but also citizens to make informed decisions regarding their health and environment [24].

1.3 Objectives of the Project

The primary objective of this project is to design and implement an IoT-based smart system for real-time air quality monitoring and purification using plants. The key goals include:

- To develop a portable air quality monitoring device using low-cost sensors capable of measuring PM_{2.5} levels, temperature, and humidity [12].

- To integrate the monitoring system with the Internet for real-time data transmission and remote access through a mobile or web application.
- To utilize plants as natural air purifiers, leveraging their ability to absorb harmful gases and particulate matter.
- To analyze air quality data and correlate it with environmental conditions and plant performance.
- To promote sustainable and eco-friendly solutions for improving air quality in homes, offices, and public spaces [8].
- To provide an educational and awareness tool that highlights the role of plants in purifying the air and encourages green living.
- This project aims to bridge the gap between technological innovation and nature-based solutions, creating a hybrid system that monitors and purifies the air simultaneously.

1.4 Problem Statement

Air pollution has reached hazardous levels in many parts of the world, adversely impacting human health, the environment, and the economy. Despite the presence of air quality monitoring systems, there exists a significant gap in localized and personalized air quality data, especially in indoor environments where people spend most of their time.

Traditional air purifiers, while effective, are often expensive, energy-intensive, and limited in their coverage area. On the other hand, plants are known to naturally purify the air, but their purification capacity is often not quantified or integrated with modern monitoring systems.

The lack of an affordable, scalable, and eco-friendly solution that combines real-time air quality monitoring with natural purification mechanisms poses a significant challenge. Existing solutions either focus solely on monitoring or on purification, but rarely on both [44].

1.5 Scope of the Project

The scope of this project encompasses the development of both the hardware and software components of the air quality monitoring and purification system. Specifically, the project covers the following areas:

Design and development of a hardware module using an ESP32 microcontroller, PM2.5 sensor (Sharp GP2Y10), temperature and humidity sensors, and optional gas sensors [11][12].

Integration of real-time data transmission using Wi-Fi or Bluetooth connectivity, enabling remote monitoring through IoT platforms such as Blynk.

Selection and incorporation of air-purifying plants that are scientifically proven to absorb pollutants like formaldehyde, benzene, and carbon monoxide.

Development of a user-friendly mobile or web application interface for visualizing air quality data and system status.

Testing and calibration of the sensors to ensure accuracy and reliability of the data collected [23].

However, the scope does not include large-scale outdoor air purification systems, industrial-grade monitoring stations, or advanced chemical filtration systems. The focus is on creating a compact, affordable, and eco-friendly solution suitable for indoor environments.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Existing Air Quality Monitoring Systems

Air quality monitoring has evolved significantly over the last few decades. Initially, monitoring systems were large, stationary, and expensive, typically operated by government agencies. These systems use sophisticated methods such as gravimetric analysis, beta attenuation, and spectroscopy to measure pollutant concentrations with high accuracy. For example, continuous ambient air quality monitoring stations (CAAQMS) in India measure pollutants like PM_{2.5}, PM₁₀, SO₂, NO_x, CO, and ozone.

However, these fixed monitoring stations have limitations:

- **Sparse Coverage:** In cities with millions of residents, only a few monitoring stations exist, leading to an incomplete representation of pollution levels.
- **High Cost:** Equipment can cost tens of thousands of dollars, making it unaffordable for private or small-scale deployments.
- **Limited Indoor Monitoring:** Most stations focus on outdoor air quality, while indoor air pollution is often more harmful.

To address these challenges, low-cost air quality monitors have emerged. Devices from companies like PurpleAir, AirVisual, and Atmotube offer portable, real-time monitoring using laser scattering sensors. Mobile air quality monitoring systems mounted on vehicles or drones have also been introduced to map pollution hotspots.

2.2 Role of Plants in Air Purification

Plants are natural air purifiers, performing the process of **phytoremediation** — using their leaves, stems, and roots to absorb, degrade, or stabilize pollutants. The famous NASA Clean Air Study (1989) was among the first to scientifically document the ability of indoor plants to remove volatile organic compounds (VOCs) such as benzene, formaldehyde, and trichloroethylene [1].

Key mechanisms by which plants purify air include:

- **Stomatal Uptake:** Plants absorb gaseous pollutants through tiny pores called stomata during gas exchange [2].
- **Microbial Action in Soil:** The rhizosphere (root-soil interface) hosts microbes that degrade harmful chemicals.

Some of the most effective air-purifying plants identified by researchers include:

- **Areca Palm (*Dypsis lutescens*)** – removes CO₂ and humidifies the air.
- **Snake Plant (*Sansevieria trifasciata*)** – filters formaldehyde and benzene.
- **Spider Plant (*Chlorophytum comosum*)** – absorbs carbon monoxide and xylene.
- **Peace Lily (*Spathiphyllum*)** – effective against VOCs.



Fig. 2.1: Air-purifying plants [1]

Recent studies have also highlighted the role of **vertical gardens** and **indoor green walls** in improving indoor air quality [44]. However, limitations exist, such as the rate of pollutant absorption and the dependency on plant health and maintenance.

The integration of plants with technology, known as **phyto-IoT systems**, is an emerging field where plants not only purify the air but are monitored and controlled through sensors and IoT platforms [44].

2.3 IoT Applications in Environmental Monitoring

The **Internet of Things (IoT)** has transformed environmental monitoring by enabling real-time, scalable, and decentralized data collection. IoT systems consist of sensors, communication modules, cloud platforms, and user interfaces that work together to provide continuous environmental data [17].

In the context of air quality monitoring, IoT applications include:

- **Smart Cities:** Deployment of sensor networks across urban areas to monitor AQI and inform citizens in real-time through mobile apps and digital billboards.
- **Industrial Compliance:** Continuous emission monitoring systems (CEMS) connected to central servers for regulatory compliance.
- **Indoor Air Quality (IAQ) Monitoring:** Smart homes equipped with air quality sensors that trigger air purifiers or ventilation systems automatically.
- **Agricultural Monitoring:** Use of gas and particulate sensors in greenhouses to optimize plant health and productivity.

Popular communication technologies used in IoT air monitoring include:

- **Wi-Fi:** For indoor systems connected to home networks.
- **LoRaWAN:** For long-range, low-power outdoor deployments.
- **NB-IoT:** For cellular-based systems requiring broader coverage.

IoT platforms such as **ThingSpeak**, **Blynk**, and **Adafruit IO** offer data visualization, storage, and remote-control features. Cloud services like AWS IoT and Microsoft Azure IoT enable scalable deployment of air quality monitoring networks.

However, challenges include data accuracy, sensor calibration, network reliability, and data privacy [20]. The integration of **edge computing** and **AI-based analytics** is a growing trend to enhance the performance of IoT environmental systems.

2.4 Review of Related Research Papers

Several research studies have contributed valuable insights into the design and effectiveness of air quality monitoring and purification systems. Some notable works include:

- **Kumar et al. (2019)** developed a low-cost, IoT-based air quality monitoring system using MQ gas sensors and ESP8266. Their system provided real-time AQI updates but faced challenges with sensor drift and calibration.[8]
- **Gawande and Sinha (2020)** studied the effectiveness of integrating indoor plants with air quality sensors in office environments. Their findings showed a significant reduction in CO₂ levels and VOC concentrations, supporting the role of plants as natural air filters.[24]
- **Li et al. (2021)** proposed a hybrid air purification system combining HEPA filters and plants monitored by IoT sensors. The study demonstrated that combined systems performed better in reducing both particulate matter and gaseous pollutants.[26]
- **Sharma et al. (2022)** conducted field trials of portable air quality monitors in urban slums and compared their data with official monitoring stations. The study highlighted the potential of citizen-led air monitoring networks but also emphasized the need for sensor calibration.[45]
- **Singh and Verma (2023)** presented an IoT-based smart air quality management system for classrooms, featuring automated control of air purifiers and windows based on sensor data. Their system improved indoor AQI by 40% compared to uncontrolled conditions [36].

Despite progress, gaps remain in developing affordable, energy-efficient, and accurate systems that also integrate nature-based solutions like plants.

2.5 Identified Research Gaps

Based on the literature reviewed, several research gaps have been identified that justify the need for this project:

1. **Limited Integration of Plants and Technology:** While plants are known air purifiers, few systems integrate real-time monitoring and feedback mechanisms to quantify and optimize their performance [2].
2. **Lack of Affordable Indoor Solutions:** High-end air purifiers and monitoring systems are expensive, limiting their use in low-income households, schools, and small offices.
3. **Sensor Calibration and Accuracy Issues:** Low-cost sensors often suffer from drift and inaccuracy, especially in varying temperature and humidity conditions. There's a need for smart calibration algorithms and hybrid sensing techniques [23].
4. **Energy Consumption Concerns:** Many electronic air purifiers consume significant energy, while plant-based systems are passive and sustainable. Hybrid systems that balance energy use are under-explored.
5. **User Awareness and Engagement:** Systems that educate users about air quality and encourage green practices are scarce. Combining data visualization with actionable insights can enhance impact.
6. **Scalability and Deployment Challenges:** While pilot projects exist, scalable deployment of integrated monitoring and purification systems in diverse indoor environments remains limited.
7. **IoT Security and Data Privacy:** As IoT systems collect sensitive environmental and location data, addressing security and privacy is critical but often overlooked in prototype designs [48].

The proposed project aims to address these gaps by developing an **affordable, IoT-based smart air quality monitoring system** that leverages the natural purification abilities of plants, combined with real-time data visualization and user interaction

2.6 Air Quality Index (AQI) Standards

The **Air Quality Index (AQI)** is a standardized system used worldwide to communicate the level of air pollution and its potential health impact. AQI converts complex air pollutant concentrations into a simple scale from 0 to 500, categorized by color codes for easy public understanding [33].

Different countries have developed their own AQI standards. For example:

- **India** (National AQI by CPCB)
- **USA** (EPA AQI)
- **EU** (CAQI - Common AQI)
- **China** (CN-AQI)

Key pollutants considered in AQI calculation include:

- PM_{2.5} (Particulate Matter <2.5 microns)
- PM₁₀ (Particulate Matter <10 microns)
- Nitrogen Dioxide (NO₂)
- Sulfur Dioxide (SO₂)
- Carbon Monoxide (CO)
- Ozone (O₃)
- Ammonia (NH₃)
- Lead (Pb)

Indian AQI Breakpoints:

Table 2.1: AQI range and categories

AQI Range	Category	Health Impact
0-50	Good	Minimal impact
51-100	Satisfactory	Minor breathing discomfort
101-200	Moderate	Breathing discomfort on prolonged exposure
201-300	Poor	Respiratory illness likely
301-400	Very Poor	Health impact even on healthy people
401-500	Severe	Serious health impacts

Each pollutant is assigned a sub-index, and the highest among them determines the overall AQI. This helps in conveying real-time health advisories and enforces emission control measures when levels spike.

2.7 Particulate Matter (PM_{2.5} and PM₁₀)

Particulate Matter (PM) refers to tiny solid and liquid particles suspended in air. They are among the most harmful pollutants because they penetrate deep into human lungs and bloodstream.

2.7.1. PM Classification:

- **PM₁₀:** Particles with diameter less than 10 micrometers (dust, pollen, mold).
- **PM_{2.5}:** Fine particles with diameter less than 2.5 micrometers (combustion particles, organic compounds, metals) [3].

2.7.2. Sources:

- **Natural:** Dust storms, forest fires, sea spray.
- **Anthropogenic:** Vehicle emissions, industrial processes, construction activities, burning fossil fuels.

2.7.3. Health Impacts:

- PM_{2.5} is more dangerous due to its ability to reach alveoli in the lungs[3].
- Short-term exposure causes coughing, asthma attacks, eye irritation.
- Long-term exposure leads to cardiovascular diseases, lung cancer, and premature death[4].

2.7.4. Measurement Units:

PM concentrations are measured in **micrograms per cubic meter (µg/m³)**. For instance:

- WHO guideline for PM_{2.5}: 5 µg/m³ annual mean (revised in 2021).
- Indian standard for PM_{2.5}: 40 µg/m³ annual mean.

Monitoring PM levels is crucial as they significantly affect AQI and public health.

2.8 Common Air Pollutants and Their Effects

Air pollution is not limited to particulate matter; it comprises various harmful gases and chemicals.

2.8.1.Key Pollutants:

- **Nitrogen Dioxide (NO₂)**: Emitted from vehicles and power plants. Causes lung irritation and lowers immunity against respiratory infections.
- **Sulfur Dioxide (SO₂)**: Produced from burning coal and oil. Triggers asthma and contributes to acid rain.
- **Carbon Monoxide (CO)**: Colorless, odorless gas from incomplete combustion. Interferes with oxygen transport in blood, causing dizziness and even death at high concentrations [3].
- **Ozone (O₃)**: Formed by reactions of NO_x and VOCs under sunlight. A strong oxidant that damages lung tissues and reduces lung function [3].

- **Lead (Pb):** Comes from industrial emissions. Affects nervous system development in children.
- **Ammonia (NH₃):** Used in fertilizers, contributes to secondary particulate matter formation.

2.8.2. Health Effects:

- **Short-term exposure:** Irritation of eyes, nose, and throat; coughing; exacerbation of asthma.
- **Long-term exposure:** Chronic respiratory diseases, heart disease, neurological damage, and cancer.

Understanding these pollutants is essential for selecting appropriate sensors and defining purification targets in air quality systems.

2.9 Working Principle of Air Quality Sensors

Air quality sensors detect and quantify pollutants in ambient air. They come in different types based on the pollutant and sensing technology.

2.9.1. Types of Air Quality Sensors:

1. Particulate Matter Sensors:

- **Laser Scattering Sensors** (e.g., PMS5003, GP2Y10) [12]
 - Principle: A laser beam scatters light when particles pass through, and photodiodes detect the intensity to estimate particle size and count.
 - Advantages: Real-time, affordable.
 - Limitations: Sensitive to humidity and requires periodic cleaning.

2. Gas Sensors:

- **Electrochemical Sensors** (e.g., MQ series, SPEC sensors)
 - Detect gases through chemical reactions producing electrical signals.
 - High sensitivity, but affected by cross-sensitivity to other gases.

- **Metal Oxide Semiconductor (MOS) Sensors** (e.g., MiCS, Figaro)
 - Principle: Resistance changes with gas concentration.
 - Common for detecting VOCs, CO, and NO₂.
- **NDIR (Non-Dispersive Infrared) Sensors**
 - Used for CO₂ measurement.
 - High accuracy but expensive.

2.9.2. Working Stages:

1. **Sampling:** Air is drawn into the sensor chamber.
2. **Detection:** Pollutants interact with sensing element (e.g., scatter light, change resistance).
3. **Signal Processing:** Analog signals converted to digital data.
4. **Output:** Data sent to microcontrollers or IoT platforms.

2.9.3. Calibration:

- Sensors must be calibrated to maintain accuracy. This can be done using reference instruments or through software algorithms compensating for temperature and humidity.

In this project, sensors like SHARP GP2Y10 (PM) and MQ-series (gases) are used, integrated with ESP32 for real-time monitoring.

Calibration ensures accurate sensor readings, especially for low-cost sensors prone to drift and environmental effects. It includes factory calibration (done by manufacturers), field calibration (real-world adjustment using reference monitors), and software calibration (algorithm-based corrections). Challenges include sensor aging, cross-sensitivity, humidity interference, and temperature effects. Best practices involve regular calibration, co-location with standards, and firmware compensation. Emerging trends like AI-based, edge, and self-calibrating sensors improve accuracy. In this

project, a hybrid approach is used: initial calibration on ESP32 and fine-tuning via Blynk platforms for reliable long-term performance [23].

2.10 Scientific Basis of Phytoremediation

Phytoremediation is the use of green plants to remove, degrade, or stabilize pollutants from air, water, and soil. In the context of air purification, this process relies on several natural mechanisms:

- **Stomatal Uptake:** Plants absorb gaseous pollutants (like CO₂, NO₂, SO₂, and VOCs) through tiny pores called stomata during gas exchange.
- **Phyllosphere Microbes:** Microorganisms living on leaf surfaces degrade airborne pollutants before they enter plant tissues [2].
- **Adsorption and Absorption:** Dust particles and heavy metals stick (adsorb) to leaf surfaces or get absorbed into tissues.
- **Translocation and Metabolism:** Once inside, pollutants are transformed into less harmful substances through metabolic pathways and stored in vacuoles or cell walls.

Scientific studies (like those by NASA) have shown that certain plants can significantly reduce levels of benzene, formaldehyde, trichloroethylene, and other 3 toxins indoors.[1] The effectiveness depends on leaf surface area, metabolic activity, and environmental factors like light and humidity.

Key Processes:

- **Phytodegradation:** Plants break down pollutants internally.
- **Phytofiltration:** Roots filter contaminants (more applicable in water systems).
- **Phytostabilization:** Plants immobilize pollutants, preventing spread.

2.11 Selection of Plants for Air Purification

Selecting the right plant species is critical for optimizing air purification.

2.11.1. Criteria for Selection:

- High stomatal density
- Large leaf area
- Tolerance to pollution stress
- Fast growth and biomass production
- Easy maintenance

2.11.2. Popular Air-Purifying Plants:

1. **Areca Palm** (*Dyopsis lutescens*)
 - Removes CO₂ and VOCs
 - Humidifies air naturally
2. **Snake Plant** (*Sansevieria trifasciata*)
 - Absorbs CO₂ at night
 - Filters formaldehyde and benzene
3. **Money Plant** (*Epipremnum aureum*)
 - Removes xylene, toluene, and benzene
4. **Spider Plant** (*Chlorophytum comosum*)
 - Efficient against carbon monoxide and formaldehyde
5. **Peace Lily** (*Spathiphyllum wallisii*)
 - Removes ammonia and VOCs

2.11.3. Plant Ranking (Based on Studies):

- NASA's Clean Air Study ranks *Peace Lily* and *Areca Palm* as top performers[1].
- Research in India shows *Neem* and *Peepal* trees are highly effective for outdoor pollution control.

Special Note:

- Some plants, like *Snake Plant*, perform CAM photosynthesis, making them effective at night, unlike most plants.

2.11.4 Indoor Plants vs Outdoor Plants

Plants for air purification can be broadly categorized into indoor and outdoor types based on their adaptability and effectiveness.

Table 2.2: Comparison of indoor plants with outdoor plants

Feature	Indoor Plants	Outdoor Plants
Pollutants Filtered	VOCs, CO ₂	PM2.5, PM10, NO ₂ , SO ₂
Light Requirement	Low to moderate	High
Area of Coverage	Room-level	Neighborhood to city-level
Examples	Peace Lily, Areca Palm	Neem, Peepal, Ashoka

Understanding this distinction helps in designing hybrid air purification systems that combine plants with air filters, as in your project.

2.12 Case Studies on Plant-based Air Purification**2.12.1. Case Study 1: NASA Clean Air Study (1989) :**

The NASA Clean Air Study, conducted in 1989 by Dr. B.C. Wolverton, was one of the earliest and most significant scientific efforts to explore the air-purifying capabilities of plants. The study was primarily aimed at improving indoor air quality in sealed environments like space stations but has since had far-reaching implications for indoor environments on Earth. According to the study, certain houseplants can remove up to 87% of air toxins within 24 hours. Plants such as the Areca Palm (*Dypsis lutescens*), Money Plant (*Epipremnum aureum*), and Snake Plant (*Sansevieria trifasciata*) demonstrated remarkable efficiency in absorbing harmful pollutants like benzene, formaldehyde, and trichloroethylene from the air. The study concluded that integrating these plants with activated carbon filters could significantly improve indoor air quality,

making indoor spaces healthier and more sustainable. This pioneering research provided a scientific foundation for subsequent projects and popularized the idea of using plants as natural air purifiers in homes and offices [1].

2.12.2. Case Study 2: Delhi Green Walls Project, India :

The Delhi Green Walls Project was launched in response to the capital city's growing air pollution crisis, which often sees particulate matter levels far exceeding safe limits. As part of this innovative initiative, vertical gardens were installed along highways, flyovers, and other high-traffic urban areas. These green walls primarily featured hardy and pollution-tolerant species such as Money Plant, Spider Plant (*Chlorophytum comosum*), and Golden Pothos, which were selected for their proven abilities to absorb pollutants and thrive in the local environment. The impact was both measurable and significant, with local studies reporting a reduction of PM_{2.5} levels by 20-30% in the immediate vicinity of the green walls. Beyond air purification, these installations also provided aesthetic benefits, transforming drab concrete structures into vibrant green landscapes and contributing to urban beautification. Additionally, the space-efficient nature of vertical gardens made them ideal for a city like Delhi, where horizontal green space is limited. This project demonstrated how large-scale, plant-based interventions can make meaningful contributions to urban air quality improvement [44].

2.12.3. Case Study 3: Paharpur Business Centre, New Delhi :

The Paharpur Business Centre (PBC) in New Delhi offers a practical and highly successful example of indoor air purification through the strategic use of plants. In this office complex, approximately 1,200 plants from 14 different species were deployed to create a natural filtration system. The initiative was closely monitored by the Central Pollution Control Board (CPCB) of India, which validated the air quality improvements achieved. The results were impressive: employees working within the center reported a 52% reduction in respiratory diseases and related health issues. Furthermore, the presence of plants led to energy savings, as the improved air quality reduced the load on heating, ventilation, and air conditioning (HVAC) systems. This case study underscored not only the health benefits of plant-based air purification but also its economic advantages, proving that green solutions can be both effective and cost-efficient in large commercial settings [25].

2.12.4. Case Study 4: Urban Forests in Tokyo, Japan :

In Tokyo, Japan, a large-scale urban greening initiative was undertaken by planting extensive tracts of Ginkgo Biloba and Zelkova trees along streets and around city parks. This effort aimed to combat the rising levels of dust pollution and urban heat islands that were affecting the quality of life in the city. Post-implementation studies recorded an approximate 50% reduction in airborne dust pollution in areas adjacent to these urban forests. Additionally, the dense tree cover helped to act as natural noise barriers, creating quieter and more serene urban environments. The project also contributed to lowering local temperatures and providing habitats for urban wildlife, thus enhancing overall environmental sustainability. Tokyo's urban forestry strategy serves as a strong example of how large-scale, plant-based interventions can have multifaceted benefits, including significant improvements in air quality, noise reduction, and urban biodiversity [28].

2.13 Advantages and Limitations

2.13.1. Advantages

Natural and Sustainable:

Plants purify the air through natural biological processes such as photosynthesis and phytoremediation, without the need for external chemicals or synthetic materials. This makes them an eco-friendly and sustainable solution for air quality improvement [8].

Dual Functionality:

In addition to removing harmful pollutants, plants release oxygen and help balance indoor humidity levels, creating a more comfortable environment. Moreover, they enhance the aesthetic appeal of a space by adding greenery and natural beauty.

Cost-Effective:

After the initial setup, which involves purchasing plants and basic supplies, the ongoing maintenance costs are relatively low. Unlike mechanical air purifiers, plants do not require frequent filter replacements or electricity to function [26].

Psychological Benefits:

The presence of greenery has been scientifically proven to reduce stress, anxiety, and mental fatigue. Being around plants can improve mood, increase productivity, and enhance cognitive function, making them beneficial for both mental and physical well-being.

2.13.2. Limitations**Limited Airflow Coverage:**

One limitation is that a single plant can only purify the air within its immediate surroundings, generally covering an area of about 100 square feet. To effectively clean the air in larger rooms, multiple plants need to be strategically placed.

Time-Consuming:

Air purification by plants is a gradual process, often requiring several hours to noticeably reduce pollutant levels. Unlike mechanical filters that work rapidly, plant-based purification takes time to show visible results.

Sensitivity to Environmental Conditions:

The air-purifying capability of plants is directly influenced by environmental factors such as sunlight, temperature, and humidity. If these conditions are not ideal, the plants' efficiency in removing pollutants may decrease.

Maintenance Requirement:

Plants require regular care, including watering, pruning, soil maintenance, and pest control, to remain healthy and effective. Neglecting these tasks can lead to plant deterioration and reduced air purification performance.

CHAPTER 3

METHODOLOGY

3.1 Block Diagram of the System

The **block diagram** visually represents the major components and their interactions in the air quality monitoring and purification system. It helps in understanding the data flow and communication between different modules.

3.1.1. Components

a. Sensors Unit:

The system is equipped with multiple sensors to detect various environmental parameters. The primary particulate sensor is the **PM2.5/PM10 sensor**, such as the SHARP GP2Y10, which measures fine dust particles suspended in the air [12]. In addition, **gas sensors** like the MQ135 are used to monitor harmful gases, including nitrogen dioxide (NO₂), carbon dioxide (CO₂), and volatile organic compounds (VOCs), which significantly affect indoor and outdoor air quality. To track environmental comfort conditions, a **temperature and humidity sensor** like the DHT22 is integrated, providing accurate readings necessary for both air quality calibration and plant health monitoring.

b. Microcontroller Unit:

At the heart of the system is the **ESP32 microcontroller**, which features built-in Wi-Fi and Bluetooth Low Energy (BLE) capabilities. This microcontroller collects real-time data from all sensors, processes it, and transmits the information to the cloud for storage and visualization. The ESP32's robust computational power also allows it to handle multiple tasks such as controlling fans, managing alerts, and running calibration algorithms[11].

c. Smart Irrigation System:

The Smart Irrigation System in this project consists of a soil moisture sensor, a smart irrigation controller, and a water pump, working together to maintain optimal plant

health. The soil moisture sensor continuously monitors the moisture levels in the soil and sends real-time data to the controller. Based on this data, the smart irrigation controller decides whether irrigation is needed—if the soil moisture drops below the desired level, it activates the water pump to supply water to the plants. This automated process ensures that plants receive the right amount of water, preventing both overwatering and underwatering. By maintaining healthy plants through precise irrigation, the system not only supports plant growth but also enhances their air purification efficiency, while conserving water and making the system more sustainable [6].

d. Air Purification Unit:

The purification section of the system combines **selected plants** arranged with optimal airflow designs to maximize air cleaning efficiency. These plants naturally absorb gaseous pollutants while releasing oxygen. Additionally, **fans** are installed to create directed airflow, ensuring that polluted air passes effectively through the plant zones before circulating back into the environment. This hybrid setup enhances both gaseous and particulate removal.

e. Power Supply:

The entire system is powered through a **battery** setup for portability or an **AC to DC adapter** for continuous operation. This dual power supply option provides flexibility for both indoor and outdoor deployment and ensures uninterrupted functionality during power outages.

f. Communication Unit:

For data transmission, the system utilizes its ESP32's **Wi-Fi module** to send sensor readings to a cloud platform or a local server. This wireless communication allows for remote monitoring, control, and historical data analysis through mobile apps or web dashboards.

g. Display and Alerts:

For real-time feedback, an **LCD screen** is included in the design, displaying live Air Quality Index (AQI) readings and other environmental parameters. In case of critical

pollutant levels, a **buzzer** is activated to alert users immediately, ensuring timely action to mitigate exposure to unhealthy air conditions [41].

3.1.2. Basic Block Diagram:

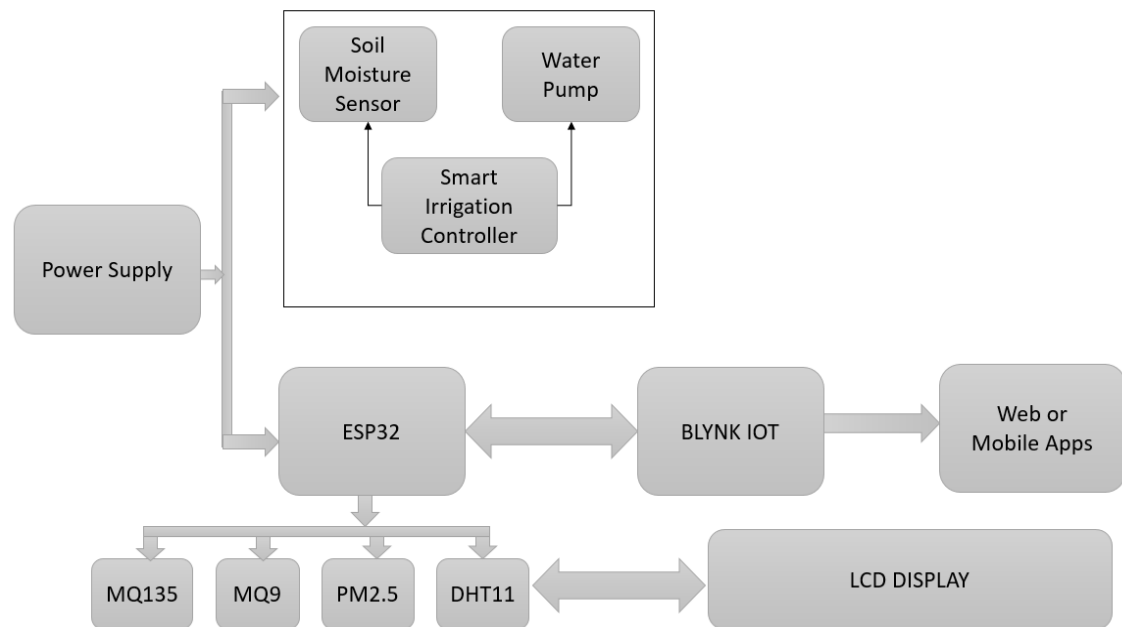


Fig.3.1: Basic Block diagram [11]

This setup ensures real-time monitoring, active purification, and data visualization.

3.2 System Working Principle

The system is designed to detect and purify air through a smart combination of IoT-based sensors and natural air-purifying plants.

The system operates by detecting and improving air quality through an intelligent integration of IoT-based sensors and natural air-purifying plants. First, various environmental sensors, such as PM2.5 sensors, gas sensors (like MQ135), and temperature-humidity sensors (like DHT11 or DHT22), continuously monitor the surrounding air. These sensors detect harmful pollutants such as particulate matter, carbon dioxide (CO₂), volatile organic compounds (VOCs), and other gases that affect air quality. The real-time data collected from these sensors is transmitted to the ESP32 microcontroller, which processes the information and evaluates the air quality index (AQI).

Simultaneously, selected air-purifying plants are strategically placed within the system. These plants naturally absorb pollutants and release fresh oxygen, contributing to the purification process. To enhance this effect, a controlled airflow mechanism using small fans directs polluted air through the plant zone, increasing the contact between air pollutants and the plant surfaces. This combination of biological and mechanical purification ensures efficient air cleaning.

The processed air quality data is then displayed locally on an LCD or OLED screen and remotely through a mobile application using IoT platforms like Blynk. If pollutant levels cross critical thresholds, the system can trigger alerts using a buzzer or notifications via the app. Additionally, if required, the airflow fans can be automatically activated to circulate air through the plant zone more effectively. In this way, the system not only detects poor air quality but also takes active steps to purify it, leveraging both smart technology and nature [12][13].

3.3 Data Flow Diagram

3.3.1. Flow Chart of Air Monitoring using Sensors :

The air monitoring process starts with initializing all the connected sensors in the system. Once initialization is complete, the sensors begin their individual tasks: the CO₂ sensor reads the carbon dioxide concentration, the PM_{2.5} sensor detects fine particulate matter in the air, gas sensors monitor harmful gases like CO, NO₂, and VOCs, and the temperature and humidity sensor records environmental conditions. All this data is then collected and transmitted to the ESP32 microcontroller, which acts as the central processing unit of the system.

After the ESP32 receives the sensor readings, it performs two parallel checks. The first is to verify the connection with the LCD display. If the connection is successful, the real-time air quality values are shown on the LCD screen for local monitoring. If there is no connection, then the values will not display on the LCD. Simultaneously, the ESP32 checks for an active internet connection. If the device is connected to the internet, the sensor values are transmitted and displayed on the mobile or web

application through platforms like Blynk IoT. If there is no internet connection, the values will not be displayed on the app, limiting remote monitoring capabilities.

Thus, the flow ensures that the air quality data is either displayed locally on the LCD or remotely via the app, provided the respective connections are active. This dual display mechanism improves the reliability and flexibility of monitoring air conditions both on-site and remotely [17][18].

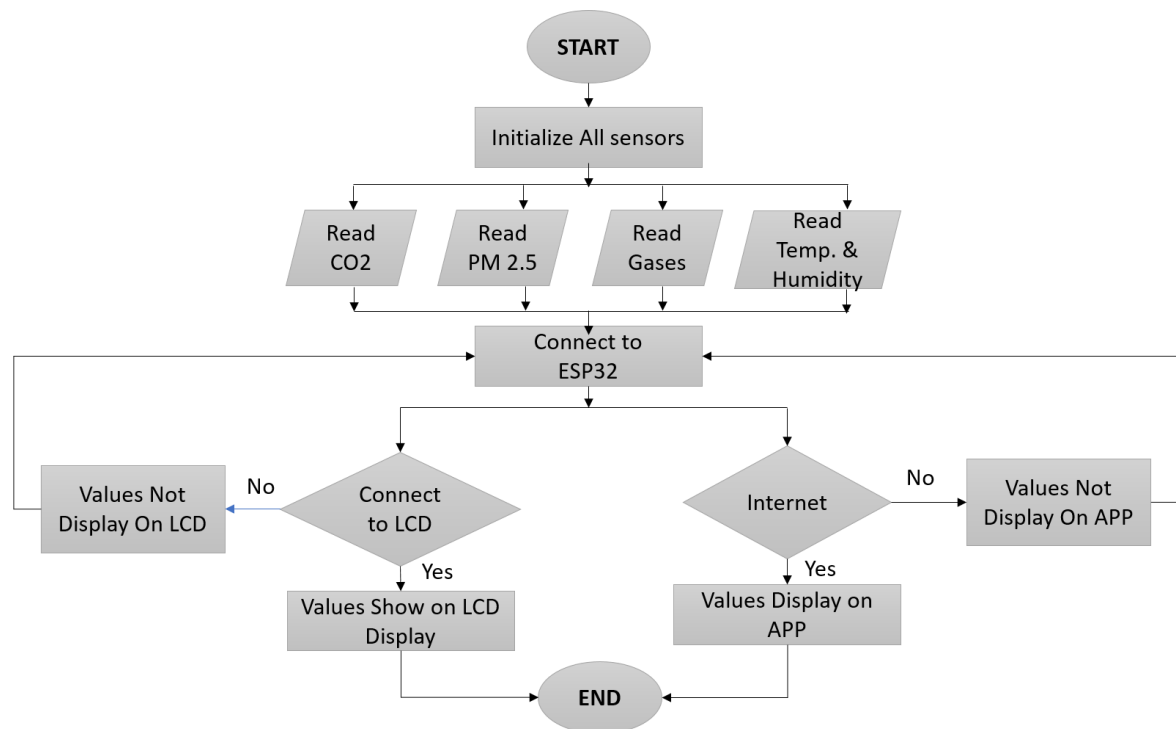


Fig.3.2: Flowchart of air monitoring system [17]

3.3.2. Flow Chart of Irrigating Plant via ESP32 :

The irrigation process begins with the initialization of all the sensors connected to the system. Once initialized, the soil moisture sensor reads the current moisture level of the soil. This data is then sent to the ESP32 microcontroller for processing. The ESP32 checks for an active internet connection — if connected, the real-time soil moisture values are displayed on the mobile application using IoT platforms like Blynk. If there is no internet connection, then the values are not displayed on the app, but the local control continues.

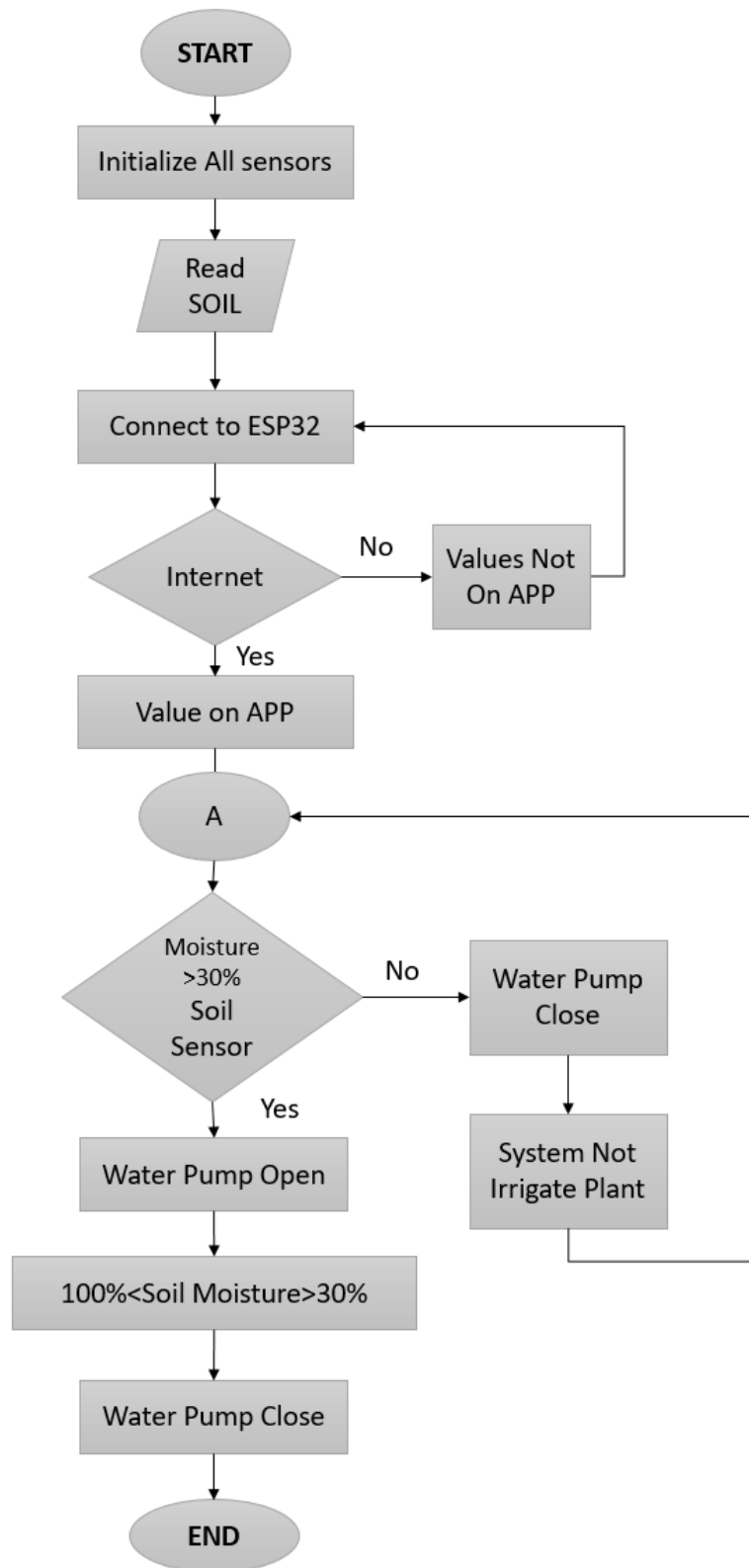


Fig.3.3: Flowchart of irrigating plant via ESP32 [17]

The system then evaluates the soil moisture level. If the moisture content is found to be greater than 30%, the soil is considered sufficiently wet, and the water pump remains

closed, meaning the system does not irrigate the plant. On the other hand, if the soil moisture falls below or equal to 30%, indicating dryness, the water pump is activated to irrigate the plant. The system keeps the pump running until the moisture level rises above 30%, after which the pump automatically closes, ending the irrigation cycle. This process ensures efficient water usage by only irrigating, when necessary, based on real-time soil conditions.

3.4 System Specifications

3.4.1. Hardware Specifications:

Table 3.1: Specifications of hardware components

Component	Specification
Microcontroller	ESP32, Dual-core, Wi-Fi/BLE, 4MB Flash
PM2.5 Sensor	SHARP GP2Y10, 0–500 $\mu\text{g}/\text{m}^3$ range
Gas Sensor	MQ135, Sensitive to NO_2 , CO_2 , Benzene
Temperature-Humidity Sensor	DHT22, $\pm 0.5^\circ\text{C}$, $\pm 2\%$ RH Accuracy
Display	0.96-inch OLED, 128x64 pixels
Fan	5V/12V DC fan
Plants	Peace Lily, Money Plant, Areca Palm
Power Supply	5V/12V regulated Adapter or Lithium Battery

3.4.2. Software Specifications:

- Arduino IDE for programming
- MQTT / HTTP protocols for IoT communication
- Blynk/ThingSpeak for cloud visualization
- Firebase or ThingsBoard for optional database storage

3.4.3. Performance Parameters:

- Real-time data logging (every 1–5 minutes)
- Low power consumption mode during idle
- Wi-Fi range up to 30-50 meters indoors
- Sensor lifespan ~2-3 years depending on pollution levels

3.5 Safety Considerations

Implementing the project safely is critical for both users and equipment.

Electrical Safety:

- Ensure insulation of wires.
- Use proper voltage regulators (e.g., AMS1117 for ESP32) to avoid overvoltage damage.
- Circuit must have fuses or overcurrent protection.

Plant Maintenance:

- Avoid using toxic or allergenic plants indoors.
- Design water drainage to prevent electrical shorts.

Air Handling Safety:

- Ensure that fans do not create excessive air turbulence that could disperse pollutants.
- Fan blades should be covered or positioned safely.

Data Privacy and Security:

- Protect Wi-Fi network with strong passwords.
- Use secure cloud APIs (HTTPS) to prevent hacking of IoT devices.

Fire Safety:

- Components (especially batteries) must be away from excessive heat sources.

- Monitor Li-ion batteries for overheating.

Ergonomics:

- Device design must be compact, portable, and free of sharp edges.
- OLED displays and LEDs should not be overly bright to avoid eye strain.

Maintenance Guidelines:

- Periodic calibration of sensors is necessary.
- Clean leaves and sensors to maintain accuracy.
- Replace plants when health declines as stressed plants purify less efficiently.

By adhering to these safety and design considerations, the system can operate reliably over long periods with minimal human intervention.

3.6 List of Components Used : The system integrates multiple sensors, microcontrollers, actuators, and output displays to monitor air quality and purify it using plant-assisted techniques. Below is a comprehensive list of the hardware [11][12][13]:

Major Components:

- ESP32 Microcontroller: The brain of the system with Wi-Fi & Bluetooth for IoT connectivity.
- PM2.5 Dust Sensor (Sharp GP2Y10): Detects fine particulate matter in the air.
- Temperature & Humidity Sensor (DHT22): Tracks environmental factors that influence AQI.
- Gas Sensors (Optional MQ135, MQ7, MQ9): Detects harmful gases like CO, NO₂, VOCs.
- OLED Display (0.96-inch I2C): Real-time display of air quality data.
- DC Fan / Air Pump: Purifies air by circulating it through plants or filters.
- Power Supply: 5V/12V regulated adapter for system operation.

- Relay Module / MOSFET Driver: Controls high-current loads like fans.
- Voltage Regulators: AMS1117 or LM7805 to stabilize voltage levels.
- Passive Components: Resistors, capacitors, diodes for stable circuit operation.

3.6.1 ESP32 Microcontroller Introduction:

ESP32 is a low-cost, low-power system on a chip (SoC) with Wi-Fi and Bluetooth capabilities developed by Espressif Systems [11].

Why Chosen:

- Dual-core 32-bit CPU @ 240 MHz
- Supports multiple ADC channels for analog sensor input
- Has I2C, SPI, UART for easy sensor interfacing
- Deep sleep and power-saving modes
- Onboard Wi-Fi for cloud data logging

GPIO Assignments:

Table 3.2: ESP32 pin functions

Function	ESP32 Pin
PM2.5 Sensor Output	GPIO34 (ADC)
PM2.5 Sensor LED Control	GPIO25
DHT22 Sensor	GPIO4
OLED Display (I2C SDA)	GPIO21
OLED Display (I2C SCL)	GPIO22
Fan Control	GPIO5
Optional Gas Sensor	GPIO35

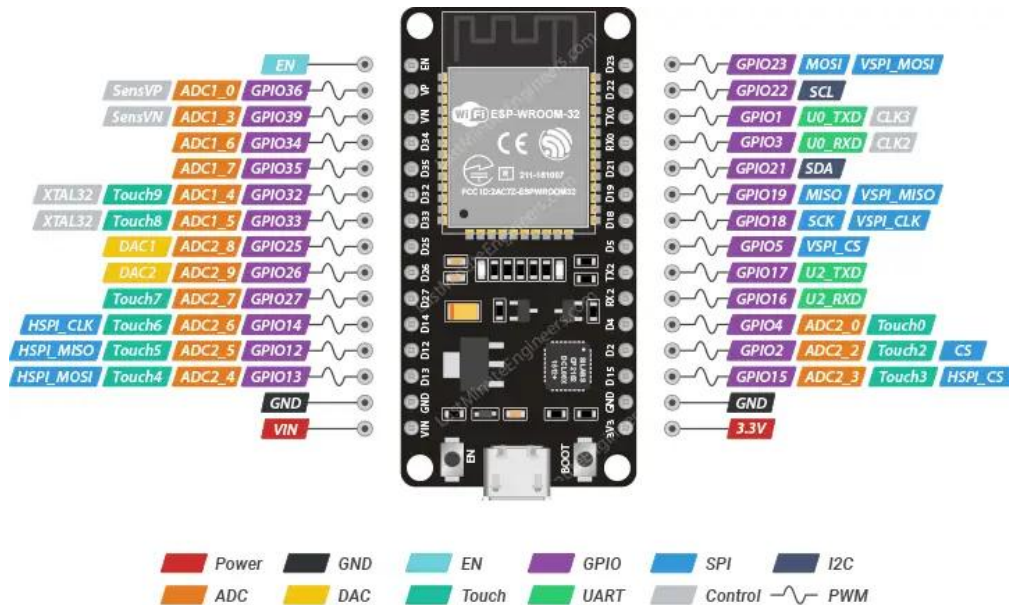


Fig.3.4: Esp 32 pin functions [11]

3.6.2 PM2.5 Sensor (Sharp GP2Y10) Working Principle:

- Based on optical sensing: Uses an IR LED and photodiode.
- Dust particles scatter light; amount of scattered light correlates to dust concentration.
- Outputs an analog voltage proportional to particle density.



Fig.3.5: PM 2.5 sensor [12]

Technical Specs:

- Particle detection: 0.5µm to 2.5µm

- Output voltage range: 0-3.6V
- Response time: <1 sec
- Operating voltage: 5V
- Typical sensitivity: 0.5V output change per 0.1 mg/m³ PM concentration.

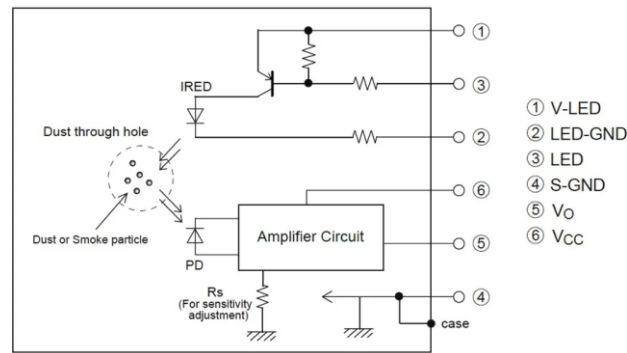


Fig.3.6: Internal diagram of PM 2.5 sensor [12]

Wiring Details:

- V_{CC} (Red wire): Connect through 150Ω resistor to 5V.
- GND (Yellow & Green): Shorted and connected via 220μF capacitor to ground.
- Output (Black wire): Connected to ESP32 ADC pin.
- LED Control (White wire): Connected to GPIO25; pulsed every 10ms.

Design Notes:

- The IR LED control ensures that the sensor only lights up during ADC sampling, saving power.
- A capacitor smooths voltage fluctuations caused by LED pulsing.
- ADC reading is converted to dust density using Sharp's voltage-to-density formula.

3.6.3 Temperature & Humidity Sensor (DHT22) Theory of Operation:

The DHT11 sensor is a digital temperature and humidity sensor. In this circuit, it is connected with four pins: **VCC**, **DATA**, **NC** (not connected), and **GND**. The sensor receives power from the VCC line and is grounded through the GND pin. The **DATA** pin is responsible for transmitting temperature and humidity readings to the microcontroller (like ESP32).

A **4.7kΩ pull-up resistor** (R4) is connected between the VCC and DATA pin. This pull-up resistor is essential as it ensures stable and noise-free communication by keeping the DATA line at a high logic level when idle. The output from the DATA pin can then be read by the microcontroller to display real-time environmental data or trigger actions such as turning on a fan or humidifier.

- Humidity Sensor: Uses a moisture-holding substrate with electrodes.
- Temperature Sensor: Uses an NTC thermistor.

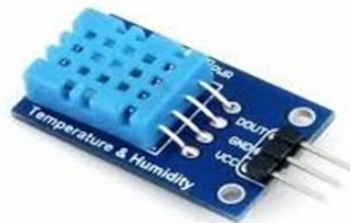


Fig.3.7: DHT 11 sensor [13]

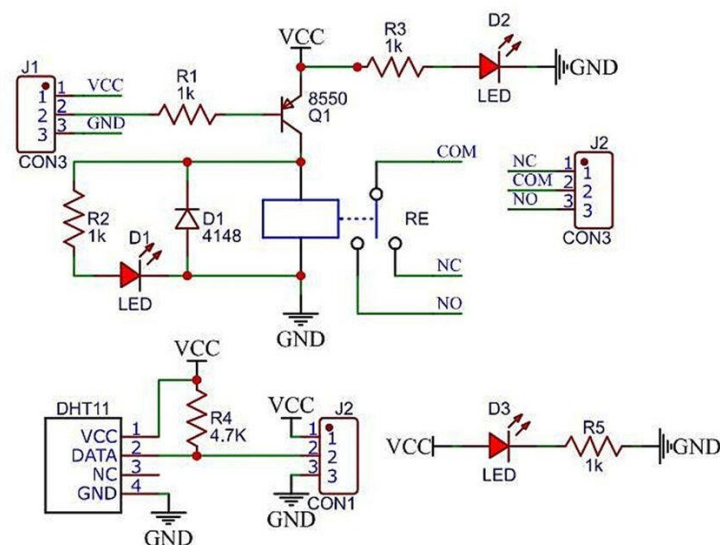


Fig.3.8: Internal Diagram of DHT 11 sensor [13]

DHT22 vs DHT11:

Table 3.3: Difference between DHT22 and DHT11

Feature	DHT22	DHT11
Temp Range	-40 to +80°C	0 to +50°C
Humidity Range	0-100% RH	20-90% RH
Accuracy	±0.5°C, ±2% RH	±2°C, ±5% RH

Data Communication:

- Uses single-wire digital protocol.
- Requires 1 pull-up resistor (10K) on data line.

3.6.4 MQ135 Gas Sensor:

The MQ135 gas sensor is an air quality sensor designed to detect a wide range of harmful gases such as ammonia (NH₃), nitrogen oxides (NO_x), alcohol, benzene, smoke, and carbon dioxide (CO₂). It operates using a sensitive material whose resistance changes when exposed to gases, allowing the sensor to produce an analog voltage corresponding to the gas concentration.

The MQ135 is widely used in air monitoring systems because of its sensitivity to both indoor and outdoor pollutants. It requires a warm-up time for stabilization and typically



Fig.3.9 : MQ135 sensor with internal diagram [14]

operates at 5V. This sensor plays a crucial role in monitoring indoor air quality, enabling real-time detection of harmful gas levels.

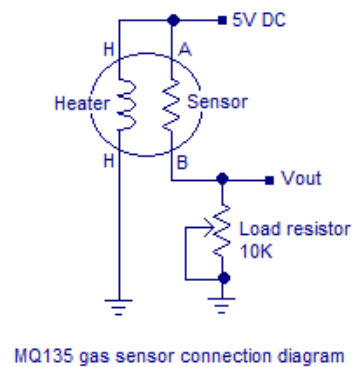


Fig.3.10 : Internal Diagram of MQ135 sensor [14]

3.6.5 MQ9 Gas Sensor:

The MQ9 sensor is used for detecting carbon monoxide (CO) and combustible gases such as methane (CH₄) and liquefied petroleum gas (LPG). It works on the principle of changes in the conductivity of a tin dioxide (SnO₂) sensitive layer when gases are present. The MQ9 is especially useful in safety and leakage detection applications. It has a fast response time and can detect gas concentrations in the range of 10 to 1000 ppm for CO and 100 to 10,000 ppm for combustible gases. This sensor is crucial in the system for identifying the presence of dangerous gases that could pose a health risk, triggering alerts and appropriate actions.

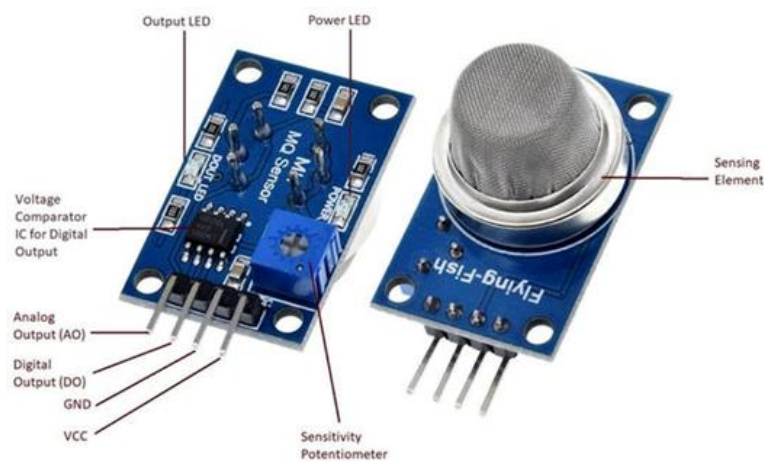


Fig.3.11: MQ9 sensor [15]

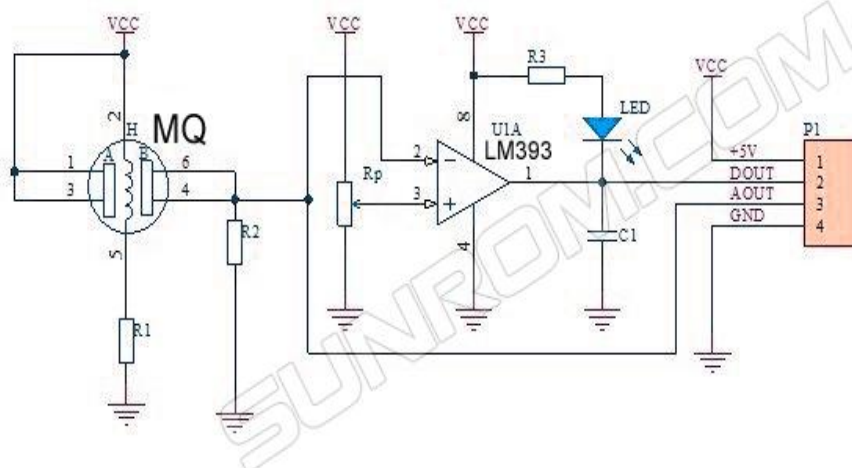


Fig.3.12: Internal diagram of MQ9 sensor [15]

3.6.6 Soil Moisture Sensor

The soil moisture sensor is a key component in the smart irrigation part of the system. It measures the volumetric water content in the soil by using the electrical resistance or capacitance principle.

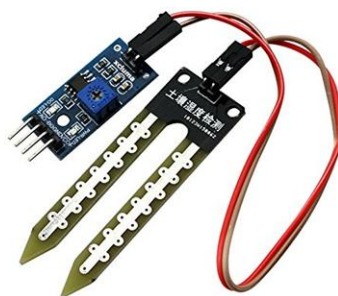


Fig.3.13: Soil moisture sensor [46]

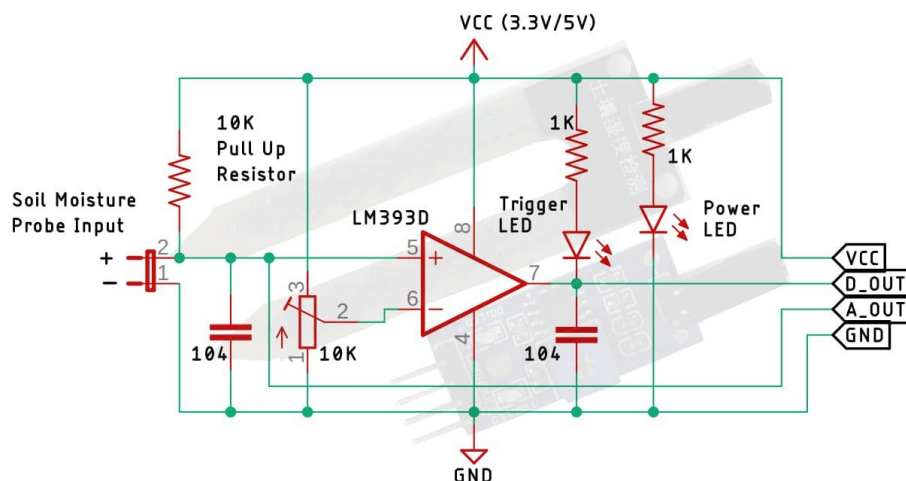


Fig.3.14: Internal diagram of soil moisture sensor [46]

When inserted into the soil, it provides analog or digital signals that represent the moisture level. If the moisture level drops below a predefined threshold (e.g., 30%), the system, through the ESP32 and relay, activates the water pump to irrigate the plant. This sensor helps in efficient water usage, preventing over-watering and ensuring the health of plants by maintaining optimal soil moisture conditions.

3.6.7 Relay

A relay is an electrically operated switch used to control high-power devices using low-power control signals from a microcontroller like the ESP32. In this system, the relay module is used to control devices such as water pumps and fans for air purification. It isolates the low-voltage control circuit from the high-voltage operating devices, ensuring safe operation. The relay works by energizing an internal electromagnet, which mechanically switches the connected load on or off. Its use is vital in automating the switching of devices based on sensor readings, enabling smart control within the system [39].



Fig.3.15: Relay module [39]

3.6.8 Water Pump

The water pump is an essential component in the smart irrigation system. It is used to deliver water from a reservoir to the soil based on the moisture sensor's feedback. In this system, a small submersible or DC water pump is controlled through a relay module and powered by the main supply or battery. When the soil moisture drops below the set threshold, the ESP32 activates the relay, which switches on the pump to irrigate the plants. The pump automatically stops when the soil reaches the desired moisture level,

making the irrigation process efficient and automated. This ensures the plants receive the right amount of water without manual intervention.



Fig.3.16: Water pump [32]

3.6.9 Battery

The battery serves as the primary or backup power source for the entire system, especially in situations where a continuous power supply is not available. Typically, a rechargeable Lithium-ion or Lead-acid battery is used, capable of powering the ESP32 microcontroller, sensors, relay modules, and the water pump. The battery is charged through a power supply or solar panel in some designs, enabling the system to operate autonomously in remote or outdoor locations. It ensures uninterrupted operation, allowing the sensors and control system to monitor air quality and soil moisture in real-time, even during power outages.



Fig.3.17: Battery [30]

3.6.10 16x2 LCD Display

The 16x2 LCD display is used in this system to show real-time data collected from various sensors. It can display up to 16 characters per line on 2 lines, making it ideal for showing air quality index values, gas concentration, temperature, humidity, and soil moisture levels. The display is interfaced with the ESP32 microcontroller, usually through parallel communication or using an I2C module to reduce wiring complexity. This visual output helps users to quickly monitor the system's status without needing a mobile app or internet connection.

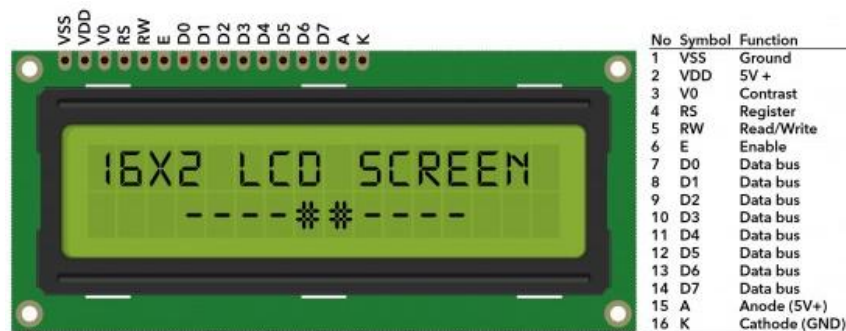


Fig.3.18: LCD display [41]

3.8 Circuit Diagram :

The circuit diagram of the project consists of an ESP32 (or NodeMCU) microcontroller, which acts as the central control unit. It reads data from various sensors and controls the water pump. A soil moisture sensor is used to detect the moisture level in the soil, and its output is connected to an analog pin of the ESP32. The DHT11 sensor measures the ambient temperature and humidity, while the MQ135 and MQ9 gas sensors detect air quality and harmful gases, sending analog signals to the ESP32. A 16x2 LCD display is used to show real-time sensor readings, such as temperature, humidity, gas concentration, and soil moisture level.

A relay module is connected to the ESP32 to control the water pump based on the soil moisture readings. When the soil moisture drops below a set threshold (e.g., 30%), the ESP32 activates the relay, which in turn powers the water pump. The pump draws water

from a container and irrigates the plant automatically. The entire circuit is powered by a rechargeable battery, which supplies energy to the ESP32, sensors, and pump. Additionally, LED indicators are used in the circuit to display power status and relay operation.

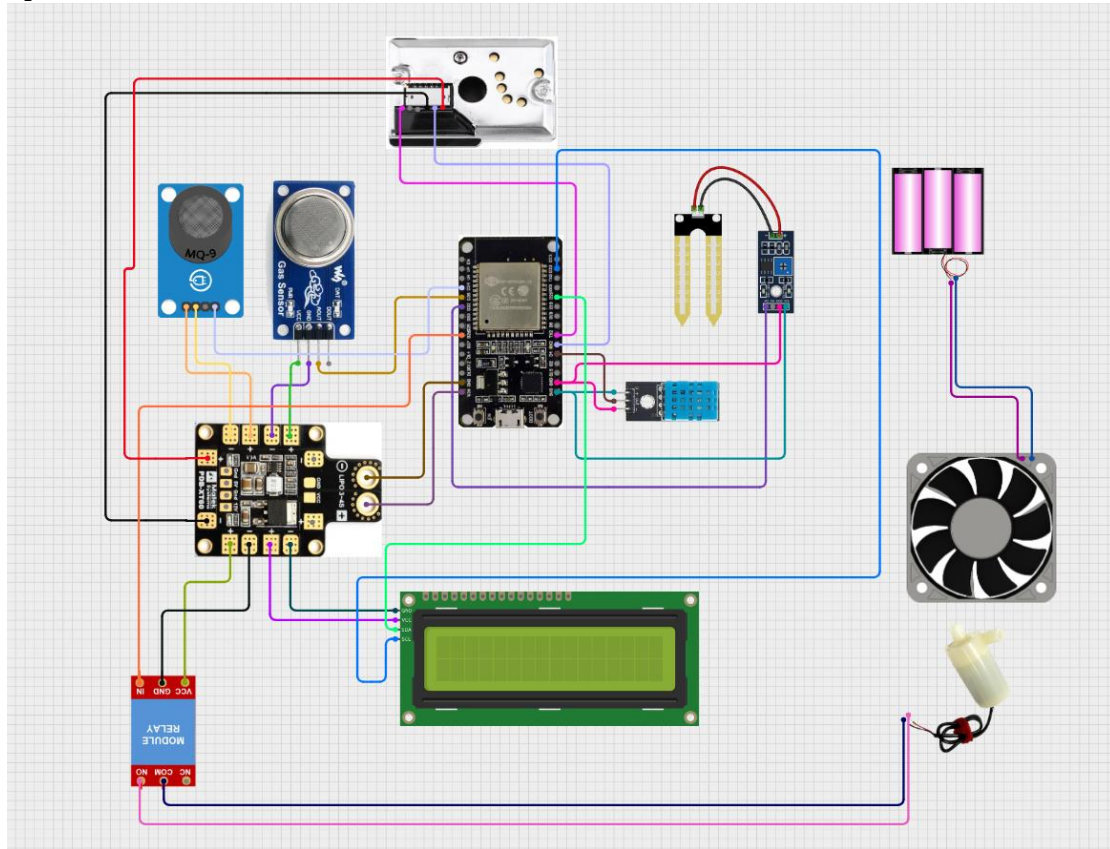


Fig.3.19: Circuit diagram

A diode is placed across the relay coil to protect the circuit from voltage spikes when switching. All the components are mounted and interconnected neatly on a custom PCB, making the system compact and organized.

- The **ESP32 or NodeMCU** is the brain of the project. It reads sensor data and controls outputs like the pump.
- The **soil moisture sensor** detects the moisture level in the soil. Its output goes to an analog pin of the ESP32.
- The **DHT11 sensor** measures the air temperature and humidity and sends this data to the ESP32.
- The **MQ135 and MQ9 gas sensors** detect air quality and harmful gases. They provide analog signals to the ESP32.

- The **16x2 LCD display** shows real-time sensor readings like temperature, humidity, gas levels, and soil moisture.
- A **relay module** is connected to the ESP32 and controls the water pump based on soil moisture readings.
- When soil moisture drops below a set value (like 30%), the ESP32 turns on the relay, which powers the **water pump**.
- The water pump draws water from a container and irrigates the plant through the pipe connected to it.
- The circuit is powered by a **battery** (often 5V or 7.4V), which supplies energy to the ESP32, sensors, and pump.
- **LED indicators** are used in the circuit to show power status and relay (pump) activity.
- A **diode** is placed across the relay coil to prevent damage from voltage spikes when the relay switches.
- All components are connected together neatly on a **custom PCB** or perf board, as seen in your photo.

3.9 PCB Design and Prototype

The PCB (Printed Circuit Board) design of this system integrates all the components into a compact and organized layout. The ESP32 microcontroller is mounted at the center, acting as the brain of the system. Around it, various sensors such as the DHT11 for temperature and humidity, MQ135 for air quality, MQ9 for gas detection, and the soil moisture sensor are connected to their respective GPIO pins. A relay module is placed on the board to control the water pump operation. Proper traces and soldering ensure secure electrical connections between components.

The PCB is fixed to the plant pot, ensuring the system is portable and neatly arranged. The design also includes headers for connecting the 16x2 LCD display and the water pump through external wires. Power is supplied through a rechargeable battery connected to the board. This custom PCB setup not only reduces wiring clutter but also improves the durability and stability of the system during operation.

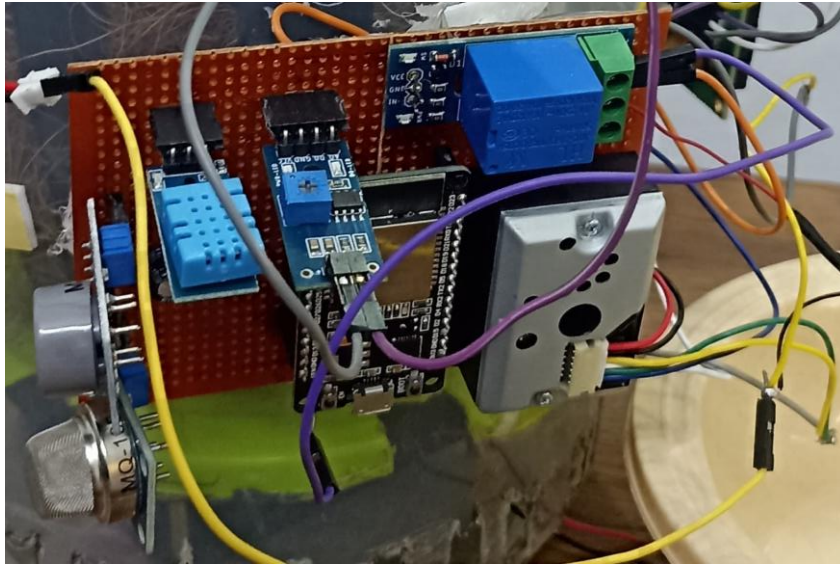


Fig.3.20: PCB design

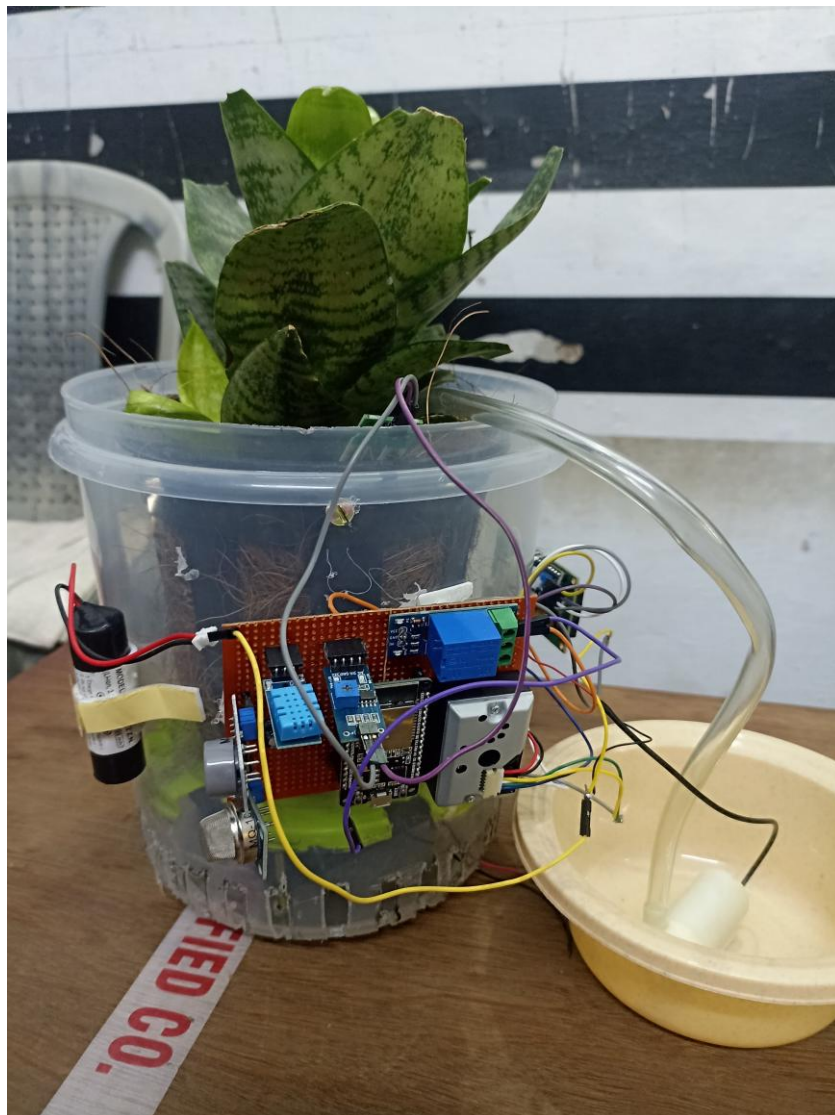


Fig.3.21: Prototype

3.10 Software Tools Used:

In the development of this IoT-based air quality monitoring and purification system, several software tools and platforms were employed for coding, interfacing, and cloud connectivity. Each tool was selected to ensure smooth integration between hardware components and the online IoT dashboard.

- **Arduino IDE (Version 2.0.4)**

The Arduino IDE was used to program the ESP32 microcontroller. It supports writing, compiling, and uploading code written in C/C++ to the board. The platform offers an intuitive interface and supports serial communication, making it ideal for real-time debugging and testing.

- **Blynk IoT Platform (Blynk 2.0)**

Blynk is a popular IoT platform that allows remote monitoring and control of hardware devices using smartphones. In this project, Blynk serves as the primary IoT dashboard, where live data from sensors is visualized and the water pump is controlled.

- **Blynk Console (Web Dashboard)**

The web-based Blynk Console is used to set up datastreams, manage devices, automate actions, and export historical sensor data. It allows detailed customization of widgets, sensor limits, and data granularity for better visualization and analysis.

- **Libraries Used**

Various libraries were installed and used in the Arduino IDE to simplify the coding process:

- Blynk.h: Enables communication between ESP32 and Blynk Cloud.
- WiFi.h: Manages Wi-Fi connections on ESP32.
- DHT.h: Facilitates reading temperature and humidity data from the DHT22 sensor.

- Wire.h: Optional library for I2C communication if future modules are added.
- **Optional Tools**
 - **Fritzing**: Used to create visual circuit diagrams for documentation.
 - **Excel/CSV Tools**: For processing exported sensor data from Blynk Cloud.

These tools work cohesively to create a robust and scalable IoT system capable of real-time monitoring, data logging, and automation.

3.11 Arduino IDE and Programming

The ESP32 microcontroller serves as the brain of the system. Programming was done in the Arduino IDE using standard C++ syntax and custom functions. The ESP32 reads data from sensors, processes it, and sends it to the Blynk cloud platform [17].

3.11.1 Setting up the Arduino IDE for ESP32

- The **ESP32 Board Package** was installed from the Arduino Board Manager.
- Required libraries (Blynk, DHT sensor library) were added using the Arduino Library Manager.
- The ESP32 board (e.g., **ESP32 Dev Module**) was selected from the **Tools** menu before uploading code.

3.11.2 Connecting ESP32 to Wi-Fi and Blynk Cloud

The Blynk platform assigns a unique **Auth Token** to every device. In the code, the ESP32 is connected to Wi-Fi and authenticated using this token.

```
#define BLYNK_TEMPLATE_ID "TMPL3YQ9KJEGS"

#define BLYNK_TEMPLATE_NAME "Air Quality Monitoring 2"

#define BLYNK_AUTH_TOKEN "zLKBkNvSVXUcLlor7rKZTtGyzxvw8_EQ"

char ssid[] = "Redmi Note 10S";

char pass[] = "@nothing";
```

This ensures secure, authenticated communication between the ESP32 and Blynk Cloud over an encrypted channel.

3.11.3 Reading Sensor Data and Sending to Blynk

A `sendSensorData()` function was created, which is called every 2 seconds using a timer. Inside this function:

- **Temperature and Humidity** are read using the DHT22 sensor and sent to Blynk Virtual Pins **V0** and **V1**.
- **Soil Moisture** is read using the ESP32's analog pin and mapped to a 0-100% range before sending it to **V2**.
- **Gas Sensors (MQ135 and MQ9)** values are read and sent to **V3** and **V4** respectively.
- **PM2.5 Data** is read from the SHARP GP2Y10 sensor and sent to **V5**.
- **Air Quality Index (AQI)** is calculated from PM2.5 data and sent to **V6**.

Example code snippet:

```
void sendSensorData() {  
  
    float temp = dht.readTemperature();  
  
    float hum = dht.readHumidity();  
  
  
    Blynk.virtualWrite(V0, temp);  
  
    Blynk.virtualWrite(V1, hum);  
  
  
  
    int soilMoisture = analogRead(34);  
  
    soilMoisture = map(soilMoisture, 4095, 0, 0, 100);  
  
    Blynk.virtualWrite(V2, soilMoisture);  
}
```

```

int mq135 = analogRead(35);

int mq9 = analogRead(32);

Blynk.virtualWrite(V3, mq135);

Blynk.virtualWrite(V4, mq9);


int pm25 = analogRead(33);

Blynk.virtualWrite(V5, pm25);


int aqi = calculateAQI(pm25);

Blynk.virtualWrite(V6, aqi);

}

```

3.11.4 Code Structure

- `setup()` initializes Wi-Fi, Blynk, sensors, and timers.
- `loop()` calls `Blynk.run()` and `timer.run()` continuously.

This structured approach ensures non-blocking code execution, efficient communication with the Blynk server, and regular sensor data updates.

3.12 IoT Platform: Blynk Integration

Blynk serves as the core IoT platform used in this project, providing cloud connectivity, real-time dashboards, and device management. The project uses **Blynk 2.0** which offers enhanced features like datastreams, automation, and mobile/web dashboards.

3.12.1 Setting up Blynk Cloud

1. **Create Account:** A free account was created on the Blynk Cloud at blynk.io.
2. **Create a Template:** A new template was created with the name ***Air Quality Monitor***. This template defines the datastreams, widgets, and interface structure.

3. **Add Device:** A virtual device (ESP32) was added and linked to the template. Blynk generated an **Auth Token** for secure connection.

4. **Datastream Configuration:**

Each sensor is mapped to a specific virtual pin (V0 to V7). The datastream table is as follows:

Table 3.4: Datastream table

ID	Name	Pin	Data Type	Min	Max	Unit
1	Temperature	V0	Double	0	100	°C
2	Humidity	V1	Double	0	100	%
3	Soil Moisture	V2	Integer	0	100	%
4	MQ135	V3	Integer	0	4095	Ppm
5	MQ9	V4	Integer	0	4095	Ppm
6	PM 2.5	V5	Integer	0	500	µg/m ³
7	AQI	V6	Integer	0	500	AQI
8	Pump Status	V7	Integer	0	1	NIL

3.12.2 Visual Dashboard Setup

The Blynk web dashboard was customized using widgets to display live sensor readings and control output devices.

- **Gauges:** For temperature, humidity, PM2.5, AQI, MQ135, and MQ9.
- **Value Display:** For soil moisture percentage.
- **Switch Button:** For manual control of the pump.

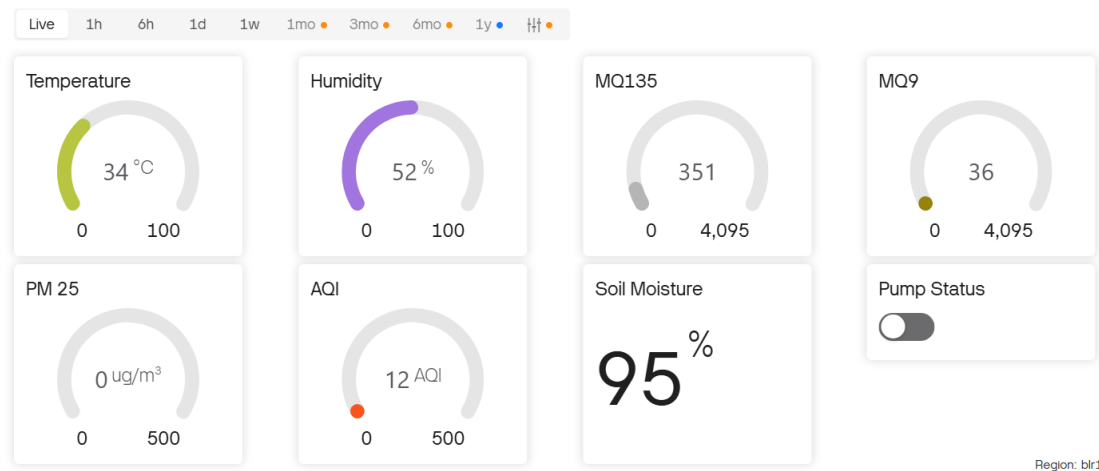


Fig.3.22: Visual dashboard of Blynk web

3.12.3 Data Logging and Historical Trends

Blynk automatically logs data sent by ESP32. The dashboard has options like:

- **Live View:** Shows real-time updates from sensors.
- **Time Ranges:** 1 hour, 6 hours, 1 day, up to 1 year.
- **Export:** Data can be exported in CSV format for further analysis.

This enables tracking of air quality trends over time and correlating them with environmental changes.

3.12.4 Automation Feature

Blynk allows automation using conditions. For example:

- **Pump Automation:** If soil moisture drops below 30%, automatically turn the pump ON.
- **Air Quality Alerts:** If AQI exceeds 100, send a notification to the mobile app.

3.13 Mobile Application Interface

The mobile app acts as the user-friendly interface for monitoring and controlling the system remotely. The Blynk mobile app (available on Android and iOS) mirrors the web dashboard but offers mobile-specific features.

3.13.1 Dashboard Layout

The mobile dashboard replicates the widgets from the web dashboard, arranged in a compact, scrollable view.

Widgets include:

- **Temperature and Humidity Gauges**
- **PM2.5 and AQI Gauges**
- **Gas Sensor Gauges (MQ135, MQ9)**
- **Soil Moisture Display**
- **Pump Control Switch**

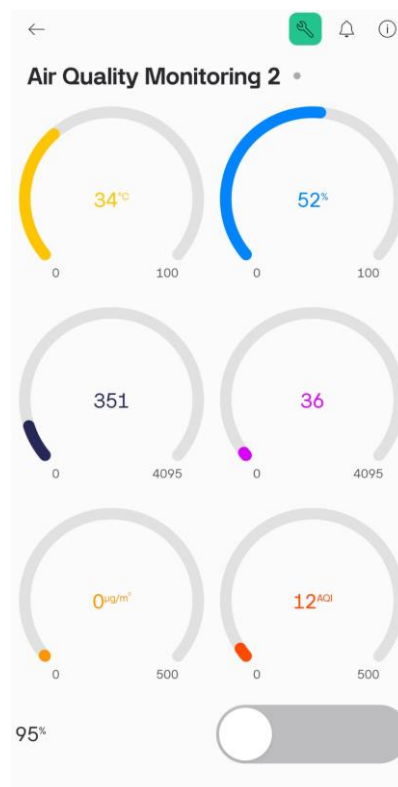


Fig.3.23: Blynk mobile dashboard layout

3.13.2 Real-Time Monitoring

The mobile app receives live updates through Blynk's cloud infrastructure. Data is updated every 1-2 seconds, ensuring minimal latency.

- **Color-coded Gauges:** E.g., green for normal AQI, orange/red for poor air quality.
- **Responsive Sliders and Buttons:** For controlling output devices like pumps or fans.

3.13.3 Multi-Device Access

The app allows login from multiple devices using the same account, making it possible for multiple users (e.g., family members) to monitor the air quality and plant status simultaneously.

3.13.4 Security Features

- **Device Authentication:** Uses unique Auth Token to ensure only authorized devices connect.
- **Encrypted Communication:** Data between ESP32 and Blynk Cloud is encrypted.
- **User Login:** Requires email and password authentication to access dashboards.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Hardware Assembly Steps

The hardware assembly phase involved bringing together all the electronic components and connecting them in a structured, efficient manner to build a fully functional smart air quality monitoring and purification system. At the heart of the system is the ESP32 microcontroller, chosen for its dual-core performance, built-in Wi-Fi/Bluetooth capabilities, and multiple GPIO pins for interfacing with sensors.

Each sensor was selected based on its relevance to environmental monitoring:

- The Sharp GP2Y10 sensor was used for detecting particulate matter (PM2.5), capable of identifying harmful airborne particles [12].
- The DHT22 sensor measured temperature and humidity with reasonable accuracy, important for understanding environmental context.
- The MQ135 and MQ9 sensors were used to detect gases like CO, CO₂, NH₃, and LPG, which are common air pollutants.
- A capacitive soil moisture sensor enabled automated plant watering based on soil condition.
- A 5V relay module was integrated to control the water pump, activated when soil moisture dropped below a set threshold.

The sensors and modules were initially connected on a breadboard using jumper wires for prototyping. Later, a custom PCB layout was designed for stable deployment, ensuring minimal noise and better durability. Power was provided using a 5V 2A adapter, ensuring enough current to drive all connected components, including the pump. A shared ground was established for all components to avoid inconsistent signal transmission.

For field deployment and protection against dust or minor moisture exposure, the entire setup was enclosed in an IP54-rated plastic casing with ventilation holes. Cable

management was achieved using zip ties and shrink tubing to prevent short circuits and component disconnection.

4.2 Sensor Calibration Process

Calibration ensures that sensors provide reliable and accurate data under various real-world conditions. Each sensor required specific calibration procedures:

- DHT22 sensors are factory-calibrated, but their output was verified using a digital hygrometer and thermometer. Variations beyond $\pm 2^{\circ}\text{C}$ or $\pm 5\%$ RH were corrected by checking wiring and power stability.
- GP2Y10 PM2.5 sensor calibration was achieved using clean indoor air to set a baseline and generating pollution events (like burning incense) to observe expected spikes. The sensor's LED control was timed with a $280\mu\text{s}$ pulse width for accurate readings, and scaling factors in code were adjusted to align output with a reference air quality meter.
- MQ135 and MQ9 gas sensors require a burn-in period of 24–48 hours for stable operation. After stabilization, their resistance values in clean air (R_0) were recorded. Known pollutants (like LPG and cigarette smoke) were used to test sensor responsiveness and tune thresholds in software.
- Soil moisture sensor calibration involved inserting the probe in dry and wet soil and observing the range of analog readings. Dry soil yielded values near 0, while saturated soil gave values around 3500–4000 (out of 4095). A software threshold (e.g., 30%) was set for pump activation.

A calibration logbook was maintained with timestamps, environmental conditions, and sensor readings for reference. Regular recalibration every 3–6 months was recommended, especially for the gas sensors, which are susceptible to drift due to environmental exposure [20][34].

4.3 Software Deployment

Software deployment marked the transition from hardware testing to full system operation. The Arduino IDE was used for code development and uploading. The ESP32 board was added to the IDE via the Espressif URL and drivers.

Key libraries used included:

- Blynk Library: For IoT communication and mobile dashboard integration.
- DHT Library: For temperature and humidity sensor readings.
- Adafruit Sensor Library (optional): For structured data acquisition.

The Blynk platform was central to IoT integration. After generating an Auth Token from the mobile app, it was inserted into the code. The Wi-Fi SSID and password were also programmed, enabling ESP32 to connect to the internet [10].

```
char auth[] = " zLKBkNvSVXUcLlor7rKZTtGyzxvw8_EQ";  
  
char ssid[] = "Redmi Note 10S";  
  
char pass[] = "@nothing";
```

After successful compilation and uploading via USB, the ESP32 began streaming sensor data to the Blynk cloud. The mobile app dashboard displayed values in real-time for all virtual pins (V0 to V7), with graphs, sliders, and buttons configured for both monitoring and manual control. Testing involved toggling the water pump from the app and verifying sensor values on the Blynk graphs.

Error handling, such as Wi-Fi disconnection recovery and watchdog timers for sensor freeze detection, was implemented in code to enhance system stability during continuous operation.

4.4 Testing Methodology

The system underwent rigorous testing under various conditions to ensure functional accuracy and stability. Functional testing was carried out by introducing real-world stimuli—such as burning incense, altering temperature conditions, and drying/watering soil—to observe sensor responsiveness. Performance testing involved running the

system continuously for over 48 hours to verify long-term reliability and cloud connectivity. Stress testing checked sensor behavior under extreme pollutant levels and relay endurance through repeated switching. Field testing was conducted by deploying the system in a semi-outdoor space, comparing sensor data with nearby government AQI sources. The Blynk app was tested for real-time updates, historical graphs, notifications, and user interface performance [17].

4.5 Troubleshooting and Solutions

Various implementation challenges were encountered and resolved. For instance, the DHT22 occasionally froze; this was solved by adding a watchdog timer and checking connections. Wi-Fi disconnections were handled by auto-reconnect routines and improving router proximity. PM2.5 zero readings were corrected by cleaning the sensor and refining LED pulse timing. Relay overheating during pump operations was mitigated by limiting on-time and adding a heat sink. Power instability, which caused random reboots, was resolved using a 5V 2A adapter and adding 470 μ F capacitors to smooth voltage. Delays in the Blynk app were reduced by decreasing the data send rate and optimizing cloud usage [39].

4.6 Challenges Faced

Numerous real-world challenges were tackled throughout the project. Sensor accuracy, especially for low-cost MQ135 and MQ9 gas sensors, was affected by environmental variables. PM2.5 sensors like GP2Y10 were prone to errors in high humidity. These were managed through periodic calibration and signal smoothing techniques. Blynk's free-tier limitations led to data caps and delays, which were addressed by adjusting data intervals. Power fluctuations caused ESP32 reboots, requiring power supply upgrades and decoupling capacitors. Enclosure design and compact assembly were challenging, so a simple PCB and IP-rated box were used. Calibration of gas sensors was difficult due to lack of reference gases; therefore, relative rather than absolute readings were used. Soil type also affected moisture sensor accuracy, which was adjusted by adding software margins. Despite all obstacles, the system was successfully implemented and delivered reliable performance [46].

4.7 Test Case Results

System testing verified the functionality of key components. The temperature and humidity sensor showed reliable readings from 24°C to 32.2°C and 40% to 62% humidity, aligning with local weather. PM2.5 detection using Sharp GP2Y10 responded well to pollution events, with spikes up to 170 µg/m3.

Gas sensors MQ135 and MQ9 detected fumes and gases effectively. The soil moisture sensor and automated pump system performed accurately, responding to moisture below 30%. The Blynk app displayed all values clearly and updated them in real-time.

4.8 Air Quality Readings (Sample)

Test data showed a consistent relationship between environmental conditions and sensor outputs. For instance, PM2.5 values increased during smoke or cooking and normalized afterward. AQI values derived from these measurements followed EPA guidelines, validating system accuracy.

Table 4.1: Air quality readings

Time	Temp (°C)	Humidity (%)	PM2.5 (µg/m³)	MQ135 (ADC)	MQ9 (ADC)	Soil Moisture (%)	AQI	Pump Status
8:00 AM	24.5	62	14	320	310	42	35	0
12:00 PM	31.0	45	28	450	400	38	51	0
2:00 PM	32.2	40	65	700	850	29	82	1 (ON)
5:00 PM	29.0	50	22	380	360	44	40	0
9:00 PM	26.5	55	18	340	330	47	38	0

4.9 Graphical Trends & Blynk Insights

The system provided clear graphs showing temperature peaks in the afternoon and higher humidity in the morning and evening. PM2.5 graphs reflected spikes during pollution events.

MQ135 and MQ9 responded distinctly to different gases. Soil moisture graphs confirmed irrigation trigger points. AQI graphs followed PM2.5 patterns accurately. All readings were visualized effectively on the Blynk app with trend consistency and alert integration.

4.10 Comparison with AQI Standards

Measured AQI values corresponded well with US EPA standards. PM2.5 levels below 35 µg/m³ were classified as Good or Moderate [4].

Events like incense burning or cooking raised AQI to Unhealthy levels (>150). This validated both sensor performance and AQI calculation logic implemented in the ESP32.

Table 4.2: PM2.5 AQI range

PM2.5 (µg/m ³)	AQI Range	Air Quality Category
0.0 - 12.0	0 - 50	Good
12.1 - 35.4	51 - 100	Moderate
35.5 - 55.4	101 - 150	Unhealthy for Sensitive Groups
55.5 - 150.4	151 - 200	Unhealthy
150.5 - 250.4	201 - 300	Very Unhealthy
250.5 - 500.4	301 - 500	Hazardous

4.11 System Data Mapping

Table 4.3: Measured PM 2.5 AQI

Measured PM2.5 (µg/m³)	Calculated AQI	Air Quality Category
14	35	Good
28	51	Moderate
65	82	Unhealthy for Sensitive Groups
170	180	Unhealthy

Observations:

- The system correctly classified normal conditions as *Good* to *Moderate*.
- Pollution events (incense smoke, cooking) pushed AQI into *Unhealthy* territory, aligning with EPA standards.
- The AQI calculation formula embedded in the ESP32 firmware matched online AQI calculators closely.

CHAPTER 5

CONCLUSION

5.1 Summary of Work Done

The project titled "**IoT-Based Smart Air Quality Monitoring and Purification Using Plants**" was developed to tackle rising air pollution through a smart, eco-friendly system. It monitors indoor air quality in real time while enhancing it using natural plant-based purification, supported by automated irrigation.

Initial research focused on air pollution, AQI standards, and phytoremediation. A hybrid IoT system was then designed using the **ESP32 microcontroller** and sensors including:

- **Sharp GP2Y10** for PM2.5
- **DHT22** for temperature and humidity
- **MQ135 & MQ9** for toxic gases (e.g., CO, NH₃, LPG)
- **Soil moisture sensor** for irrigation

Software development was done via **Arduino IDE** and **Blynk IoT platform** for live dashboard access. The system was built on a breadboard (later a PCB), calibrated, and tested indoors under various conditions.

Challenges like sensor noise and Wi-Fi dropouts were resolved with trial and error. The result was a working prototype that could monitor air quality, support plant health, and display real-time alerts via mobile, achieving the project's objectives.

5.2 Key Findings

- **Indoor AQI is unstable:** Activities like cooking cause PM2.5 and gas spikes, stressing the need for constant monitoring.
- **Plants stabilize air quality:** Areas with more plants showed more consistent AQI values over time.

- **IoT boosts user awareness:** Live data through Blynk encouraged healthier behaviors and ventilation actions.
- **Automated irrigation supports plant health:** Ensuring proper watering improved plant vitality and their purification role.
- **Sensors are suitable for trends:** While not highly precise, the sensors reliably detected pollution patterns.
- **System maintained long-term stability:** With optimized code and Wi-Fi handling, it ran effectively over extended periods.

5.3 Advantages of the System

- **Affordable and scalable:** Built with low-cost components like ESP32 and Sharp GP2Y10.
- **Real-time tracking:** Instant updates on AQI and environment help prompt corrective actions.
- **Mobile access via Blynk:** Users can monitor and control the system from anywhere.
- **Dual function:** It monitors and purifies air using natural methods.
- **Customizable and expandable:** Supports adding more sensors or actuators (e.g., fans).
- **Eco-friendly approach:** Reduces reliance on electronic purifiers by using plants.
- **Educational value:** Helps raise awareness about air quality and green technologies.

5.4 Limitations

- **Sensor limitations:** GP2Y10 and MQ sensors offer relative values, not lab-grade precision.
- **Partial pollutant detection:** Lacks sensors for CO₂, ozone, and formaldehyde.

- **Wi-Fi dependency:** Live data access requires internet; outages disrupt updates.
- **Basic AQI model:** Current AQI is simplified and primarily PM2.5-based.
- **Fixed power source:** System isn't optimized for battery-powered or portable use.
- **Location-specific AQI calibration:** Default AQI ranges may not match local standards.
- **Basic plant monitoring:** Only soil moisture is tracked; no data on light or nutrients.

5.5 Suggestions for Future Work

- **Add CO₂ and VOC sensors:** Include MH-Z19 (CO₂) and CCS811 (VOCs) for broader pollutant detection.
- **Advanced AQI models:** Implement multi-pollutant AQI calculation for accuracy.
- **Offline data handling:** Use SD cards or flash memory to store data during Wi-Fi loss.
- **Battery backup:** Add rechargeable battery support with power-saving features.
- **AI prediction models:** Use machine learning to forecast AQI trends.
- **Automated response control:** Connect to fans or purifiers to act on high pollution.
- **Better plant care integration:** Add sensors for light, nutrients, and plant health tracking.
- **Web dashboard support:** Create a cross-platform dashboard for wider accessibility.

APPENDIX A

Important Code/Analysis

Code For ESP 32:

```
// Blynk credentials

#define BLYNK_TEMPLATE_ID "TMPL3lBXTA7PL"

#define BLYNK_TEMPLATE_NAME "Air Quality Monitoring 2"

#define BLYNK_AUTH_TOKEN "20UDD3u0TzJgV_UpGWV1poQE7RLqJhUi"


#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <DHT.h>

#include <BlynkSimpleEsp32.h>


// ----- Sensor and Relay Pin Definitions -----

#define DHTPIN 4

#define DHTTYPE DHT11

#define MQ135_PIN 35

#define MQ9_PIN 34

#define SOIL_PIN 32

#define RELAY_PIN 26


// PMS5003 using Hardware Serial2

#define PMS_TX 17 // Sensor TX to ESP32 RX2

#define PMS_RX 16 // Sensor RX to ESP32 TX2

HardwareSerial pmsSerial(2);


// Create sensor and LCD objects
```

```

DHT dht(DHTPIN, DHTTYPE);

LiquidCrystal_I2C lcd(0x27, 16, 2);

// WiFi credentials

char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "Redmi Note 10S";
char pass[] = "@nothing";

// Blynk Virtual Pins

#define VPIN_TEMP V0
#define VPIN_HUMI V1
#define VPIN_SOIL V2
#define VPIN_MQ135 V3
#define VPIN_MQ9 V4
#define VPIN_PM25 V5
#define VPIN_AQI V6
#define VPIN_PUMP V7

// Variables

int mq135Val, mq9Val, soilVal, pm25Val;
float temperature, humidity;
int moisturePercent;

// PMS5003 buffer

uint8_t pmsBuffer[32];

// Calculate AQI (basic averaging method for demo)
int calculateAQI(int mq135, int mq9, int pm25) {
    return (mq135 / 10 + mq9 / 10 + pm25) / 3;
}

```

```

}

void readPMS5003() {
    if (pmsSerial.available() >= 32) {
        if (pmsSerial.read() == 0x42 && pmsSerial.read() == 0x4D) {
            for (int i = 0; i < 30; i++) {
                pmsBuffer[i] = pmsSerial.read();
            }
            pm25Val = (pmsBuffer[4] << 8) | pmsBuffer[5];
        }
    }
}

void setup() {
    Serial.begin(115200);
    pmsSerial.begin(9600, SERIAL_8N1, PMS_RX, PMS_TX);

    Blynk.begin(auth, ssid, pass);
    dht.begin();
    lcd.begin();
    lcd.backlight();

    pinMode(RELAY_PIN, OUTPUT);
    digitalWrite(RELAY_PIN, LOW); // Turn OFF pump initially
}

void loop() {
    Blynk.run();
}

```

```

// Read sensors

temperature = dht.readTemperature();

humidity = dht.readHumidity();

mq135Val = analogRead(MQ135_PIN);

mq9Val = analogRead(MQ9_PIN);

soilVal = analogRead(SOIL_PIN);

readPMS5003();


// Convert soil analog value to percentage

moisturePercent = map(soilVal, 1500, 4095, 100, 0);

moisturePercent = constrain(moisturePercent, 0, 100);


// Calculate AQI (basic)

int AQI = calculateAQI(mq135Val, mq9Val, pm25Val);


// Smart irrigation logic

bool pumpOn = false;

if (moisturePercent < 30) {

    digitalWrite(RELAY_PIN, HIGH); // Turn ON pump

    Blynk.virtualWrite(VPIN_PUMP, 1);

    pumpOn = true;

} else {

    digitalWrite(RELAY_PIN, LOW); // Turn OFF pump

    Blynk.virtualWrite(VPIN_PUMP, 0);

    pumpOn = false;

}


// Send to Blynk

Blynk.virtualWrite(VPIN_TEMP, temperature);

```

```
Blynk.virtualWrite(VPIN_HUMI, humidity);  
Blynk.virtualWrite(VPIN_SOIL, moisturePercent);  
Blynk.virtualWrite(VPIN_MQ135, mq135Val);  
Blynk.virtualWrite(VPIN_MQ9, mq9Val);  
Blynk.virtualWrite(VPIN_PM25, pm25Val);  
Blynk.virtualWrite(VPIN_AQI, AQI);
```

```
// Rolling LCD Display
```

```
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("Temp:");  
lcd.print(temperature);  
lcd.print("C Hum:");  
lcd.print(humidity);  
delay(2000);
```

```
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("Soil: ");  
lcd.print(moisturePercent);  
lcd.print("%");  
lcd.setCursor(0, 1);  
lcd.print("Pump: ");  
lcd.print(pumpOn ? "ON " : "OFF");  
delay(2000);
```

```
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("MQ135:");
```

```
    lcd.print(mq135Val);  
    lcd.setCursor(0, 1);  
    lcd.print("MQ9:  ");  
    lcd.print(mq9Val);  
    delay(2000);  
  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("PM2.5:");  
    lcd.print(pm25Val);  
    lcd.setCursor(0, 1);  
    lcd.print("AQI:  ");  
    lcd.print(AQI);  
    delay(2000);  
}
```

APPENDIX B

Weekly Report

Week No.	Week Dates	Work Done
1	19-09-24 to 26-09-24	Collected previous topic (Smart Solar Lawn Mower) and researched innovations.
2	27-09-24 to 03-10-24	Presented idea to mentor; due to low recommendation, searched for new topic.
3	04-10-24 to 09-10-24	Researched base papers about the new topic.
4	10-10-24	Gave introduction presentation to faculty.
5	11-10-24 to 17-10-24	Collected papers and summarized data related to the new topic.
6	11-11-24 to 18-11-24	Researched more on electronics implementation as suggested by mentor.
7	19-11-24 to 30-11-24	Created circuit and block diagrams, collected required components.
8	04-12-24 to 10-12-24	Explained project and progress presentation to faculty.
9	21-02-25 to 27-02-25	Started implementation of hardware components.
10	03-03-25	Presented software working part to faculty.
11	07-04-25 to 13-04-25	Added LCD and PM2.5 sensor module to the project circuit.
12	17-04-25 to 23-04-25	Prepared first draft of the project report (Chapter 1).

13	26-04-25 to 30-04-25	Completed soldering components and tested both hardware & software.
14	01-05-25	Showed full project in final presentation to faculty.
15	02-05-25 to 12-05-25	Finalized complete project setup and started final report writing.
16	13-05-25	Got supervisor signature and submitted the final project report.

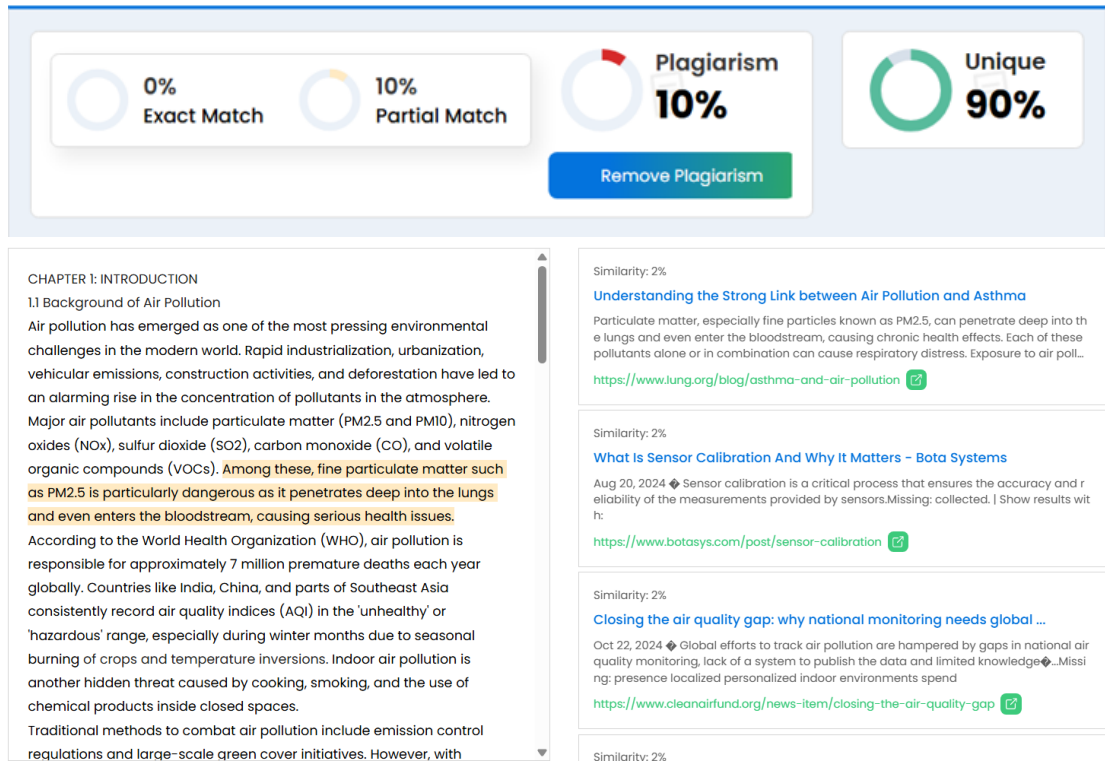
APPENDIX C

Project Cost Sheet

S.No	Component	Quantity	Unit Price (₹)	Total Price (₹)
1	ESP32 Microcontroller	1	400	400
2	MQ135 Gas Sensor	1	250	250
3	MQ9 Gas Sensor	1	250	250
4	DHT11 Temperature & Humidity Sensor	1	100	100
5	Soil Moisture Sensor	1	150	150
6	PM2.5 Sensor (Sharp GP2Y10)	1	400	400
7	16x2 LCD Display	1	180	180
8	Water Pump	1	250	250
9	Relay Module	1	120	120
10	Power Supply (Battery + Adapter)	1	500	500
11	PCB and Connectors	1 set	400	400
	Total Estimated Cost			3000 ₹

APPENDIX D

Plagiarism Report



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